

Informed Trading by Advisor Banks: Evidence from Options Holdings

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Strong conflicts of interest exist within investment banks: the investment banking division possesses substantial private information, and the asset management division seeks such information. This raises the question of whether the asset management division benefits from an information advantage on client firms. While prior examinations of advisor bank trading in client firms have focused on stocks and found mixed results, we argue that the options market represents a more attractive venue for such trading. We find significant evidence of advisor banks trading in client firm options ahead of merger announcements, but no evidence of similar trading in client firm stock. (*JEL* G14, G18, G24, G34)

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In financial markets, information is valuable: market participants with superior information can earn superior returns. In advance of major corporate events such as mergers, inside information is particularly valuable but also particularly guarded; only a limited number of people should know of the pending event. Nevertheless, dynamics in both stock and options markets provide strong

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evidence of information leakage, raising the question of how the information dissipates.¹

Advisor banks represent one of the few parties that have access to this privileged information. While strict regulations prohibit investment banking divisions from sharing client information, it is challenging to fully guard all information. To the extent that information regarding pending mergers cannot be completely safeguarded, it is an empirical question whether information dissipates more rapidly within the same bank, compared with across institutions. Because the investment banking and asset management divisions of a bank are both geographically proximate and potentially share social connections, it seems likely that at least some information leakage would occur along this channel. Prior literature provides contradictory evidence on such information leakage: Bodnaruk, Massa, and Siminov (2009) claim strong evidence that banks trade on client information, Kedia and Zhou (2014) find weak evidence, and Griffin, Shu, and Topaloglu (2012) find no evidence.

One striking aspect of the above analyses and associated contradictory findings is that they all focus on trading in the debt or equity of client firms. We argue that options represent a more powerful setting. As noted by Black (1975), options represent a less costly means to leverage private information. The advantages of options relate to the fact that they have limited downside, for example if an expected merger is withdrawn.² Consistent with these features, Easley, O'Hara, and Srinivas (1998) find that option markets are venues for informed traders, and Cao, Chen, and Griffin (2005) find that the option market displaces the stock market for information-based trading during periods immediately preceding information-intensive events, where they infer information-based trading from patterns in price and volume.

A widening in the bid-ask spreads of target firms' options prior to the merger announcement suggests an increase in informed trading around this time (see, e.g., Cao, Chen, and Griffin 2005). Dai et al. (2011) conclude that hedge funds represent a portion of this informed trading, but beyond this, relatively little is known about the identity of the institutions behind these trades. Several factors motivate advisor banks as representing one potential source of these trades. First, advisor banks are among the first to know about many pending mergers, thus increasing the likelihood that information will leak to the asset management divisions of these banks, even if inadvertently. Second, there is prior evidence of information breakdowns within investment banks—for example, the publicized

¹ See, e.g., Meulbroeck (1992), Acharya and Johnson (2010), Jayaraman, Frye, and Sabherwal (2001), Augustin, Brenner, and Subrahmanyam (2015), and Heitzman and Klasa (2017).

² See also Back (1993), Cao (1999), Cremers and Weinbaum (2010), Johnson and So (2011), Lin, Lu, and Driessen (2015), Roll, Schwartz, and Subrahmanyam (2010), and Xing, Zhang, and Zhao (2010). The theoretical advantages of options depend on the presence of capital constraints, which arguably vary across different banks. However, individual traders are likely to also face internal controls such as value at risk (VAR) measures, which require traders to define the most they could lose in dollars over the next month (or year). The limited downside of options means they have less impact on VAR measures.

2006 lawsuit in which Morgan Stanley paid \$10 million to settle charges related to such violations. A 2012 Securities and Exchange Commission (SEC) report highlights concerns related to the sharing of material nonpublic information between divisions of banks and to the lack of oversight regarding any potentially related trading. Finally, a *Wall Street Journal* article suggests that such problems may be particularly severe in the case of mergers, because stocks are frequently not put on watch lists until long after merger talks begin.³

To empirically examine the propensity of advisor banks to make informed trades as well as the form of such trading, we construct a sample of 912 mergers in which the target is public and has options that are publicly traded, and the target and acquirer advisors have trading arms that file 13F statements and trade options. From 13F filings on EDGAR, we extract the number of call option positions, the number of put option positions, and the shares of stock held in the target firm, at the end of each of the six quarters preceding the merger announcement. To best enable us to assess the information advantage of advisor banks relative to the many other market participants, we collect these data for all 13F-filing institutions, including advisor banks, non-advisor banks, hedge funds, and all other 13F-filing entities, which include insurance companies and mutual fund families.

The first portion of the paper focuses on the extent to which advisor banks trade in the client firms' options. We find that in the quarter prior to the merger announcement, advisors substantially increase their holdings of calls relative to puts in the target firm. Among target advisors, the net call ratio, defined as calls minus puts as a fraction of millions of shares outstanding, averaged 4.5% over Quarters -6 through -3, compared with 7.9% in Quarter -1. Regression analyses support these univariate patterns: advisors significantly increase their net call ratio in the target firm, and the magnitude of the increase is significantly greater than that of other investment banks and hedge funds.

Patterns in advisor banks' holdings of options in the acquirer firm provide an informative contrast. Limiting the sample to deals that will likely have a meaningful effect on the acquirer firm, defined as deals with a relative size of at least 5%, we find that advisors are significantly more likely than non-advisors to increase their holdings of puts in the acquirer firm—but not their holdings of calls. The stark differences in advisor banks' trading in target versus acquirer firms are consistent with the large positive average returns of target firms around merger announcements, compared with the small, negative average returns of acquirer firms.

Even if information leakage is more likely between divisions of the same bank, compared with across institutions, it is nevertheless unlikely that the asset management division will learn of and trade on every pending deal.

³ Watch lists are lists of potential deals on which the bank is advising; banks are required to monitor trading in the affected securities of all firms on such lists. "Trading in Deal Stocks Triggers Look at Banks," January 14, 2008, *Wall Street Journal*.

We find that the observed patterns are significantly related to the feasibility of trading options. For example, effects are greater among firms with higher options volume relative to stock volume and among advisor banks with greater participation in options markets, which is consistent with these factors relating to lower trading costs and a lower probability of trades raising suspicion among regulatory authorities. We also find that patterns are stronger among advisor banks that have had a prior advisory relationship with the target firm. Finally, we find some evidence that advisor banks with weaker governance structures—for example, with poison pills or CEO-Chair duality—are more likely to engage in this form of informed trading. In contrast, we find no evidence that deal-specific terms play a significant role. We conjecture that while advisor banks have sufficient information to assess an increased probability of a takeover, they are not privy to more specific information such as deal terms.

Further analyses support the conjecture that the asset management division's information advantage derives directly from the advisory relationship, rather than from an exogenous factor that causes certain banks to be both more informed and selected as advisor. We find no evidence that a pseudo-advisor, that is, a non-advisor bank with an ex ante high probability of being selected as advisor, engages in similarly informed trading prior to merger announcements.

Within our sample period, many banks ceased proprietary trading due to a provision of the Dodd-Frank Act commonly known as the Volcker Rule. While the primary intention of this regulation was to limit bank risk, our findings highlight the substantial effect that it had on informed trading within the bank. Specifically, following banks' adherence to the Volcker Rule, we find no evidence of the banks significantly changing options holdings ahead of client firms' merger announcements.

The last portion of the paper considers our findings relative to those in prior literature. First, Cao, Chen, and Griffin (2005) conclude that around information-intensive events, price discovery occurs in options markets before being incorporated into stock prices. This suggests a positive relation between advisors' options positions and subsequent target firm run-ups prior to the merger announcement, and our findings are consistent with this conjecture. Second, to the extent that options dominate stock as a means of trading on private information, as suggested by many theoretical models, advisors will be less likely to trade in client firm stock. Our findings are also consistent with this prediction. Consistent with Griffin, Shu, and Topaloglu (2012), we find no evidence that advisor banks are more likely than other institutions to purchase target firm stock in advance of the merger.

Our paper contributes to several streams of literature on informed trading. First, while there exists a wide literature on informed trading by company insiders, less is known about trading by the banks that advise these companies. Around key corporate events such as mergers, advisor banks have as much information as company insiders. Second, while prior papers such as Cao, Chen, and Griffin (2005), Cremers, Fodor, and Weinbaum (2017),

Augustin, Brenner, and Subrahmanyam (2015), and Augustin et al. (2017) infer informed trading around mergers based on patterns in aggregate options data, we exploit data that enables us to compare trading across institutions with varying levels of inside information. While many researchers have examined institutions' trading in stock around corporate events, to the best of our knowledge we are the first to investigate how institutions that are potentially informed trade in options markets during these periods. In sum, our analysis highlights the interplay between divisions of advisory banks and the ways in which private information affects trading in different types of securities.

1. Data

1.1 Merger sample

We obtain a sample of mergers and acquisitions between 2000 and 2016 from Thomson Financial. As is typical in many merger analyses, we focus on mergers in which the acquirer owned less than 50% of the target prior to the deal and owned 100% after the deal. Because we are interested in the options holdings by advisor banks in the target firms, we also impose several other limitations. First, we require the target to be publicly traded and to be listed on OptionMetrics for the six quarters prior to the merger announcement. Second, we require the target and acquirer firm to have used an advisor bank that served as advisor in at least ten deals throughout our sample period and that has a trading arm as identified by the filing of 13F statements.

Within our sample period, the Dodd-Frank Act represents a substantial regulatory change. Especially relevant for our study is the provision banning proprietary trading, commonly known as the Volcker Rule. Many banks ceased proprietary trading either as a direct result of this rule or in expectation of its implementation. Online Appendix A1 lists the date on which each of these banks ceased proprietary trading. In many cases these banks still engaged in market-making activities, meaning they might still hold options positions, but these positions would not be for speculative purposes. For each bank, we search through company filings and news releases to determine if it ceased its proprietary trading operations, and if so, we record the date of this change. Across the twenty-seven banks in our sample, eleven ceased proprietary trading during our sample period. Several banks—for example, Jefferies and Oppenheimer—specifically state that they are not subject to the Volcker Rule because they are not bank holding companies, so they continue to engage in proprietary trading. Mergers in which all advisor banks had ceased proprietary trading are excluded from our main sample. This is motivated by our focus on informed trading—banks subject to the Volcker Rule should not have a trading desk that focuses on such trading. In total, our main sample includes 912 mergers. For certain analyses, we contrast this sample with mergers that occurred after advisor banks' adherence to the Volcker Rule.

SEC regulations require all institutional investment managers with at least \$100 million under discretionary management to list their equity-linked holdings (including options) on 13F filings. Databases such as Thomson Financial provide only stock holdings. Therefore, we download all 13Fs and amended 13Fs directly from EDGAR. Many investment banks file 13Fs under multiple names, and we take care to fully match each advisor name to all relevant 13F filer names. We also account for bank mergers during the sample period. In cases where a bank filed an amended 13F, we are careful to obtain the last-reported holdings. If the amended filing is a restatement we pull the holdings from the 13F/A, whereas if the amended filing adds new holdings entries, we search both the 13F and the 13F/A.

EDGAR 13F filings were not reliably available on EDGAR prior to 1999 (though equity holdings can be obtained for earlier years using Thomson Financial), and it is this limitation that dictates the beginning of our sample period. The availability of 13F filings on EDGAR is depicted in the Online Appendix Figure A1, which shows the number of distinct filers with 13F reports each quarter, from 1990 to 2016.

Using PERL coding and extensive manual checking, we extract the options holdings from each of these filings. Because many of the institutions filing 13Fs never hold options, we limit our sample to institution quarters in which the institution participated in the options market, which we define as holdings options in at least one quarter over the past year. This minimum is defined across all publicly traded firms, meaning there is no requirement that they hold options in any of our sample firms. Our main specifications include investment banks that have served as advisor, as well as hedge funds. In robustness checks we also include other institutions that meet the SEC filing requirements.

We obtain deal-specific information from Thomson Financial (e.g., payment terms, etc.). For both the target and the acquirer, we use CRSP to obtain stock returns and Compustat for accounting data. We retrieve institutional stock holdings from the compiled 13F holdings available from Thomson Reuters. The choice of this source rather than the filings on EDGAR is dictated by the longer history of stock holdings data available with Thomson Reuters. To allow comparison with existing studies, we analyze stock holdings starting in 1984.

Descriptive statistics on these 912 mergers are provided in Table 1. Panel A shows target and acquirer firm characteristics, and panel B shows deal characteristics. While our mergers are similar along many dimensions to broader merger samples (see, e.g., Bodnaruk, Massa, and Siminov 2009 and Cremers, Nair, and John 2009), our requirement that the target firm have publicly listed options results in several differences, which are broadly similar to the differences documented by Barraclough et al. (2013), who impose a similar requirement. For example, target firms in our sample are larger and have higher trading volume, with an average (median) market capitalization of \$3.69 (\$1.39) billion and an average (median) trading volume of 111.17 (39.66) million shares. This is consistent with Mayhew and Mihov's (2004) findings

Table 1
Descriptive Statistics*Panel A: Target and acquirer firm characteristics*

	Target firms			Acquirer firms		
	Mean	Median	Std. dev.	Mean	Median	Std. dev.
Market capitalization (\$bil)	3.69	1.39	7.16	26.33	6.92	52.23
Market leverage	0.18	0.15	0.18	0.16	0.13	0.15
Volatility (annualized)	0.46	0.38	0.31	0.38	0.31	0.24
Option volume (thous)	142.54	17.00	617.65	512.42	105.91	1122.42
Share volume (mil)	111.17	39.66	350.08	296.50	94.61	603.75
Stock return (qtrly)	0.04	0.04	0.26	0.05	0.04	0.27
EBITDA / assets	0.09	0.11	0.16	0.12	0.13	0.12
Market to book	1.91	1.50	1.73	2.27	1.64	2.07
Cash / assets	0.19	0.10	0.21	0.17	0.10	0.18
CapEx / assets	0.05	0.03	0.06	0.05	0.03	0.06
R&D / sales	0.24	0.00	2.93	0.06	0.00	0.38

Panel B: Deal characteristics

	Mean	Median	Std. dev.
Cash merger	0.53	1	0.50
Termination fee	0.83	1	0.37
Merger rumor in Quarter -1	0.05	0	0.22
Merger rumor between end of Quarter -1 and merger announcement	0.07	0	0.26
Target and acquirer in same industry	0.68	1	0.47
Tender offer	0.19	0	0.39
Target CAR _{-1 to +1}	0.21	0.18	0.22
Acquirer CAR _{-1 to +1}	-0.02	-0.02	0.09
Target run-up	0.07	0.02	0.23
Acquirer run-up	0.02	0.00	0.15
Premium	0.32	0.28	0.42
Withdrawal	0.14	0	0.34

Panel C: Options and equity positions of advisors

	Advisor to target firm				Advisor to acquirer firm			
	Mean	Median	Std. dev.	Inter-quartile range	Mean	Median	Std. dev.	Inter-quartile range
Fraction of deals in which advisor owns calls in target firm	25.8%				25.3%			
Fraction of deals in which advisor owns puts in target firm	21.9%				22.7%			
Fraction of deals in which advisor owns equity in target firm	91.9%				95.2%			
Stock ownership ratio	0.81%	0.29%	3.45%	0.86%	0.75%	0.38%	2.98%	0.79%
Call ratio	15.84%	0.00%	101.20%	3.94%	15.57%	0.00%	100.26%	7.10%
Put ratio	9.73%	0.00%	67.03%	0.10%	10.95%	0.00%	72.76%	5.36%

This table shows statistics on the 912 mergers between 2000 and 2016, in which the acquirer owned less than 50% of the target firm prior to the merger announcement and 100% after the announcement, and the target was publicly traded and had listed options for the six quarters prior to the merger announcement. Finally, the target and acquirer advisors must have a trading division that files 13F statements and participates in the options market, defined as holding options in at least one quarter over the past year and not having ceased proprietary trading due to the Volcker Rule. (Further detail on changes in banks' proprietary trading operations due to the Volcker Rule are provided in Online Appendix Table A1.) Variable descriptions are provided in the Appendix. Mean, medians, and standard deviations are shown for all variables. Panel A shows target and acquirer firm characteristics as of the quarter end prior to the merger announcement (fiscal year end for accounting variables). Panel B shows deal characteristics. Panel C describes the options and equity positions of the target and acquirer advisors in the target firm, as of the last quarter end prior to the merger announcement.

regarding the types of stocks that exchanges tend to select for option listing. Perhaps not surprisingly, the associated acquirer firms are larger as well, with an average (median) market capitalization of \$26.33 (\$6.92) billion. We also have a slightly lower rate of withdrawal, 14%, which is likely driven by the fact that our sample is tilted toward larger target firms. For the average target firm, the quarterly trading volume equals 142,540 options (where each option traded represents a contract on 100 underlying shares of common stock). In 68% of mergers, the target and acquirer are in the same Fama-French 12-industry group, and 53% of our deals are cash mergers.

1.2 Measurement of options held by advisors

For each target firm, we use 13F data from EDGAR to aggregate the number of shares underlying call and put options that are held by each target advisor, each acquirer advisor, and each non-advisor, where non-advisors include any 13F filer not serving as advisor to the deal in question. We identify the subsample of filers that are hedge funds using the approach of Boyson, Ma, and Moordian (2016). In total, we have 27 banks (which by definition serve as advisor in at least some deals), 218 hedge funds, and 251 other 13F filers that participate in the options market during our sample period. To provide the strongest set of empirical tests, our main analyses are based on the sample of banks and hedge funds, as prior literature suggests that these are the entities that are most likely to be informed. Thus, for each deal we examine whether the advisor banks' trades are significantly different than those of non-advisors, where non-advisors include both hedge funds and other banks not advising the target or acquirer firm. In the robustness section, we consider other benchmark samples—for example, extending the benchmark sample to all 13F filers, limiting the benchmark sample to entities that never served as advisor, or defining the benchmark only across banks.

Because the 13F data are available only as of the end of each March, June, September, and December, the exact number of months between a quarter end and the merger announcement date varies. We refer to Quarter –1 as the last quarter end prior to the merger announcement, which will be between 1 and 90 calendar days prior to the merger announcement. Analogously, Quarter –2 will be between 91 and 180 days prior to the merger announcement. We track the options and stock holdings of each target firm for six quarters prior to the merger announcement.⁴ Throughout the paper, we focus primarily on options and stock held in the target firm. This focus on the target firm is motivated by the fact that the target experiences substantially larger stock price movements around acquisition announcements. For comparison, we also conduct a similar analysis on the acquirer firm.

⁴ While prior studies of options holdings, e.g., Pan and Poteshman (2006), Cremers, Fodor, and Weinbaum (2017), and Augustin, Brenner, and Subrahmanyam (2015), have employed data at finer intervals, the use of quarterly data enables us to learn the identity of the investor holding the options.

For each deal, we tabulate each institution's call options holdings, measured as the number of shares underlying call options as a fraction of total shares outstanding. Put option holdings are measured analogously. Using these data, we compute the following three measures of the institution's option holdings:

$$\text{Call Ratio} = \frac{\text{Calls}}{\text{Shares Outstanding (mil)}} \quad (1)$$

$$\text{Put Ratio} = \frac{\text{Puts}}{\text{Shares Outstanding (mil)}} \quad (2)$$

$$\text{Net Call Ratio} = \frac{\text{Calls} - \text{Puts}}{\text{Shares Outstanding (mil)}} \quad (3)$$

In addition, we also compute the institution's stock holdings at the end of each of the six quarters leading up to the merger announcement:

$$\text{Stock Ownership Ratio} = \frac{\text{Shares Held}}{\text{Shares Outstanding}} \quad (4)$$

Ratio measures are winsorized at the 99.5% level to mitigate the influence of outliers (the left-hand winsorization would be inconsequential) and expressed in percentage terms.

Panel C of Table 1 shows that as of the quarter end preceding the merger announcement, target advisors own call options, put options, and stock in 26%, 22%, and 92%, respectively, of target firms. Percentages for acquirer advisors are similar. Average equity ownership in the target firm as a percentage of market capitalization is 0.8% among target advisors and 0.75% among acquirer advisors. Finally, call (put) option holdings as a percentage of millions of shares outstanding are an average 15.8% (9.8%) among target advisors and 15.6% (10.9%) among acquirer advisors. Further detail regarding the holdings of both advisor and non-advisor institutions in the target firms across the six quarters preceding the merger announcement are provided in Online Appendix Table A2.

The asset management divisions of the 27 advisor banks in our sample are substantially larger than those of the hedge funds: median bank-quarter holdings consist of 3,058 securities, compared with 92 securities for hedge funds (not tabulated). However, hedge funds are more likely to invest in options. For the median hedge fund quarter, 7% of securities represent options, and one-quarter of hedge fund quarters have 22% or more of the portfolio in options. The analogous percentages for banks are 5% and 19%.

Institutions are instructed to report only options that they hold, and to not report put or call options that they write. Thus, short options positions are not reported on Form 13F or subtracted from a long position in the same issuer. These requirements are similar to those for stock holdings, and thus our omission of short option positions is similar to the analogous omission of short stock positions in the many studies of institutional stock holdings. While

we have the number of options held, we do not have data on the exercise price, expiration date, or dollar value of these options positions. Section 7 discusses the extent to which our conclusions regarding advisor bank-informed trading may be influenced by these data limitations.

We use RavenPack News Analytics to download all news stories about the target and acquirer firms during the six full quarters prior to the merger announcement. RavenPack extracts textual information from a wide variety of sources, including newswires, newspapers, and social media sites. Using textual analysis techniques, RavenPack structures these data into a format that is usable for quantitative research, and it has been used both in academic studies and for trading purposes. We code any news story for which the Type, Category or Sub-Type fields is “merger,” “business-comb,” or “acquisition” as being merger-related. We then code *Merger Dummy* equal to one if there was a merger-related news story during the quarter.

1.3 Institutional background

The prohibition of investment banks sharing information across divisions, what is commonly referred to as “Chinese walls,” comes under Section 15(g) of the Securities Exchange Act of 1934: “Every registered broker or dealer shall establish, maintain, and enforce written policies and procedures ... to prevent the misuse ... of material, nonpublic information.” Section 204A of the Investment Advisers Act of 1940 creates a similar obligation for investment advisors.⁵ Monitoring is conducted via both self-regulatory organizations and regulatory agencies, where the former includes the National Association of Securities Dealers (NASD), the New York Stock Exchange (NYSE), and after 2007 the Financial Industry Regulatory Authority (FINRA), and the latter includes the SEC, Department of Justice (DoJ), and the Federal Trade Commission (FTC).⁶

The SEC has continually monitored adherence to these regulations, but nevertheless reports issued by the SEC express concerns. A 2012 SEC report highlights four concerns: (i) significant interaction between groups with material nonpublic information and groups engaged in trading, (ii) senior executives considered to be “above the wall” with no monitoring or related restrictions, (iii) formal discussions in which material non-public information was provided to sales, trading, or research personnel for business purposes with no review of any subsequent trading, and (iv) “gaps in oversight coverage at

⁵ Quoting from a 1990 report by the Division of Market Regulation, U.S. SEC, “Broker-dealer Chinese Walls have evolved to include policies and physical apparatus designed to prevent the improper or unintended dissemination of market sensitive information from one division of a multi-service firm to another (i.e., from the mergers and acquisitions area to proprietary or retail trading). ... Within this very general framework utilized by the majority of multi-service broker-dealers, no two systems are alike, ranging from very tight-centralized control of information and review to little or no review or follow-up.”

⁶ See the internet appendix of Ahern (2017) for additional detail.

most broker-dealers.”⁷ Consistent with these concerns, Ahern (2017) finds that an advisor bank was the source of leaks in 23 of 285 insider trading cases related to mergers.

2. Patterns in Options Holdings

Our examination of the stock and options positions of advisors several months ahead of the merger announcement is motivated by the fact that many advisors are informed about the likelihood of a merger within this timeframe. Boone and Mulherin (2007) find that for deals initiated by the target firm, this process typically begins six to seven months prior to the actual merger announcement. In the subsample of two hundred deals examined by Dimopoulos and Sacchetto (2014), the median time between the beginning of the takeover process and the actual public merger announcement is seven months in cases initiated by the target firm, and four months in other cases. Our focus on a relatively extended period prior to the merger contrasts with most of the literature on insider trading, which has concentrated on more narrow windows around the merger announcement. Meulbrook (1992) focuses on stock trading in cases that were litigated by the SEC, finding that the insider trading occurs on average thirteen days prior to the merger announcement (the median is six days). Augustin, Brenner, and Subrahmanyam (2015) study aggregate options activity prior to merger announcements, and they find that volume is significantly higher over the thirty days prior to announcement, with patterns among SEC litigated cases similar to those in the overall sample. Acharya and Johnson (2010) similarly focus on aggregate option trading patterns, and they study only five days prior to the merger announcement.

Patterns in options markets suggest information leakage beginning thirty to sixty trading days prior to the merger announcement. Cao, Chen, and Griffin (2005) report that options volume increases thirty to fifty trading days (i.e., 1.5 to 2.5 months) prior to the merger announcement, with the increase in call options exceeding that of both puts and stock. We find similar patterns within our sample (not tabulated). Consistent with the spread containing information on the magnitude of informed trading (see, e.g., Glosten and Milgrom 1985), Cao, Chen, and Griffin (2005) find that the bid-ask spread for both calls and puts is wider over the thirty trading days prior to the merger announcement than during the benchmark period (defined as Days $-200, -100$), and we find that the spread begins to increase approximately two to three months prior to the merger announcement (not tabulated).

Figures 1 and 2 show the options holdings of parties that we conjecture may be informed—that is, the advisors to the target and acquirer firms. Figure 1 focuses on the average net call ratio across target (acquirer) advisors versus non-advisors

⁷ Staff Summary Report on Examination of Information Barriers: Broker-Dealer Practices under Section 15(g) of the Securities Exchange Act of 1934 (<https://www.sec.gov/about/offices/ocie/informationbarriers.pdf>).

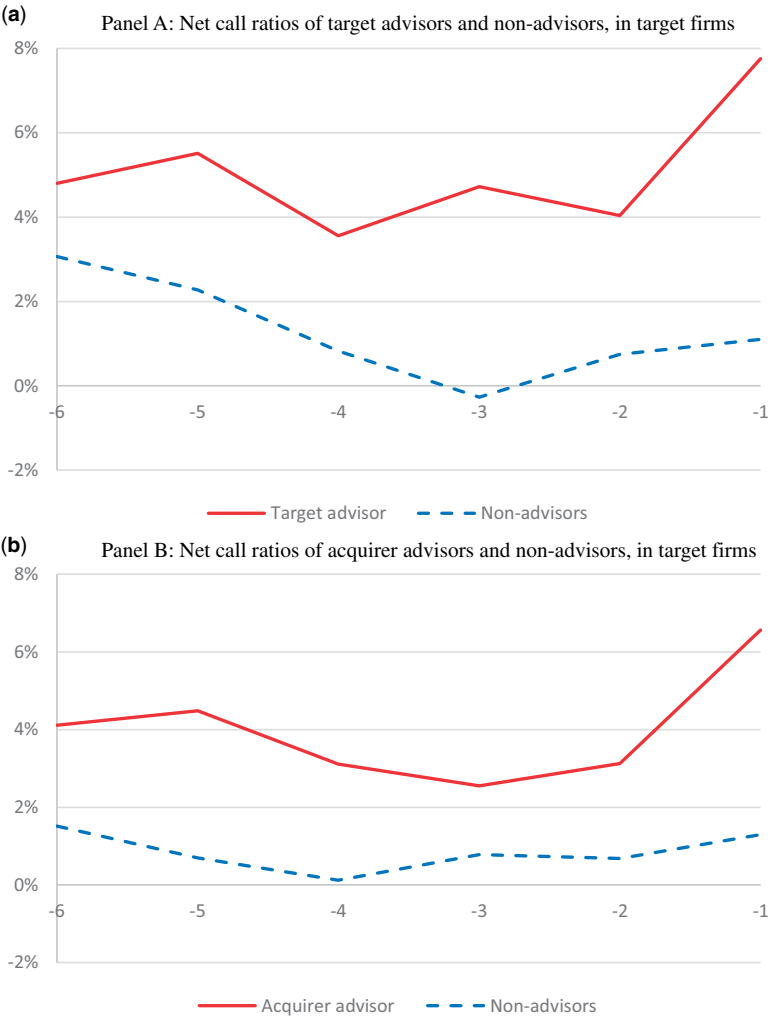
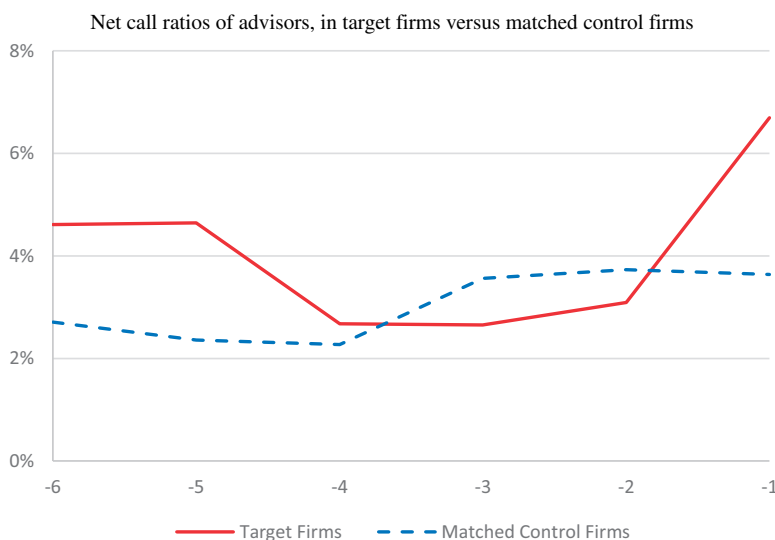


Figure 1
Option holdings of advisor versus non-advisors in the target firm, before merger announcements
The sample consists of 912 mergers between 2000 and 2016, as described in Table 1. Panel A compares target advisors' net call ratios in the target firm, measured as $\frac{Calls - Puts}{Shrs Out(mil)}$, to those of matched non-advisors. For each advisor-firm observation, we select the non-advisor with the closest net call ratio in the target firm in Quarter -6, and follow that non-advisor over the subsequent quarters. Panel B shows the difference between acquirer advisors' and non-advisors' holdings in the target firms, where non-advisors are similarly matched to acquirer advisors. For each deal, we calculate an advisor-weighted average across all deals, where more weight is given to deals with more target (panel A) or acquirer (panel B) advisors.

in panel A (panel B). For each deal, we calculate an advisor-weighted average across all deals, where more weight is given to deals with more target (panel A) or acquirer (panel B) advisors. To facilitate comparison, we benchmark each advisor to the non-advisor, either a bank or a hedge fund, with the closest net

**Figure 2****Option holdings of advisors in target versus matched control firms, before merger announcements**

The sample consists of 912 mergers between 2000 and 2016, as described in Table 1. The figure compares advisors' *Net Call Ratios* (where advisors include both target and acquirer advisors) in the target firm, measured as $\frac{\text{Calls} - \text{Puts}}{\text{ShrsOut}(\text{mil})}$, to their holdings in matched control firms. For each target firm, we identify a matched control firm based on a propensity score model. The control firm represents the non-target firm with the highest predicted probability (excluding the actual target) of being the target firm,

$$\begin{aligned} \text{Target Firm} = & \alpha + \beta_1 \log(\text{MktCap})_t + \beta_2 \text{Volatility}_{Q_{Trt}} + \beta_3 \log(\text{Options Volume})_{Q_{Trt}} + \beta_4 \log(\text{Options Volume})_{Q_{Trt}-1} \\ & + \beta_5 \text{Share Volume}_{Q_{Trt}} + \beta_6 \text{Stock Return}_{Q_{Trt}} + \beta_7 \text{Stock Return}_{Q_{Trt}-1} + \beta_8 \text{Leverage}_t \\ & + \beta_9 \text{Industry Merger Activity} + \text{Quarter FE} + \text{Fama-French 48 Industry FE} + \varepsilon \end{aligned}$$

where this regression is estimated four quarters prior to the merger announcement, and control firms are required to have publicly traded options.

call ratio in that firm during Quarter -6. Each panel shows holdings of advisors (solid line) and of non-advisors (dashed line) each quarter. Looking first at panel A, we see that six quarters prior to the merger announcement, target advisors' net call ratio is slightly higher than that of the non-advisors: 4.81% for target advisors versus 3.06% for matched non-advisors. This slight difference so far in advance of the merger likely reflects a greater inclination for the advisor bank to hold call options in the firm, for reasons not directly related to the merger. However, as the merger approaches, the difference increases markedly, from 1.7% in Quarter -6 to 6.7% in Quarter -1. Panel B of Figure 1 shows a similar though less strong pattern for acquirer advisors. The difference between the net call ratio of acquirer advisors and non-advisors increases from 2.6% in Quarter -6 to 5.3% in Quarter -1.

Figure 2 shows a comparison of advisors' options holdings in target firms versus matched control firms. We employ a propensity score approach to identify the control firms. Specifically, we estimate a probit model, where

the dependent variable equals one if the firm was an actual target, zero otherwise. Independent variables include *log(market capitalization)*, *volatility*, *log(options volume)*, *log(share volume)*, *quarterly return*, *leverage*, *industry merger activity*, and *calendar quarter* and Fama-French 48 industry fixed effects. These variables represent the same factors that we include as control variables in our main analyses and are defined in the Appendix. All variables are defined as of Quarter -4 relative to the merger announcement. We select the firm with the fitted value closest to that of the target firm as the control firm. As shown in Online Appendix Table A3, the control sample firms are largely similar to the target firms. The stock and options volume of target firms is greater than that of matched control firms, but the statistical significance is relatively low (p -value = 0.08). All other differences are insignificant at conventional levels.

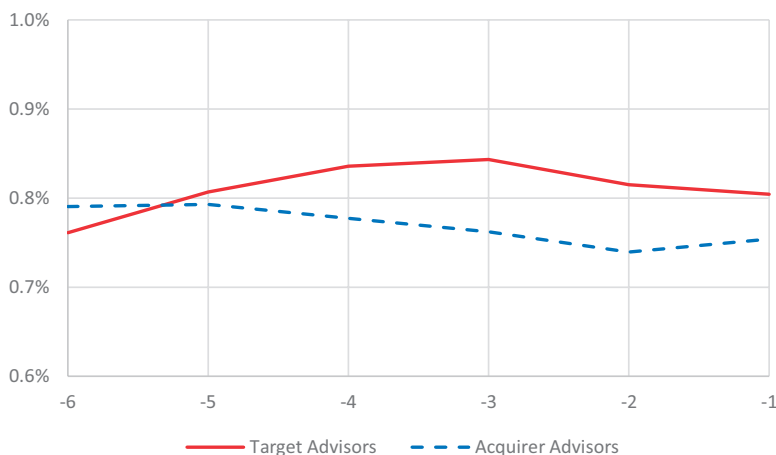
Figure 2 shows that advisors' net call ratio increases substantially more in target firms than in matched control firms, in the quarter prior to announcement. Advisors' net call ratio holdings are on average 0.9% higher in target firms in Quarters -6 through -3 , compared with 3.1% higher in Quarter -1 .

Finally, Figure 3 shows the trends in stock holdings in the target firm as a fraction of shares outstanding for target advisors and for acquirer advisors. In contrast to the options evidence presented in Figures 1 and 2, we see no evidence that advisor banks also increase their stock holdings in the target firm prior to the merger announcement. The much stronger evidence of increases in call options, compared with stock holdings, is consistent with both theoretical arguments regarding the advantages of options markets for informed traders and with the empirical evidence of Cao, Chen, and Griffin (2005) that informed trading tends to move to options markets ahead of information-intensive events.

3. Multivariate Evidence

3.1 Advisor bank trading in target firms

Tables 2–4 examine the robustness of the patterns observed in Figures 1 and 2. Table 2 investigates whether advisors' options holdings in the target firm change relative to non-advisors' holdings in the same firm as the merger approaches. The dependent variables equal *Call Ratio* in Column 1, *Put Ratio* in Column 2, and *Net Call Ratio* in Column 3. We estimate panel regressions, with the sample consisting of the 912 mergers $\times$ six quarters $\times$ the banks and hedge funds in existence at that point in time. As discussed previously, for any bank that ceased proprietary trading due to the Volcker Rule, we include the bank in the sample only until this cessation of trading. Independent variables of interest include *Quarter -1* , *Quarter -2* , and each of these dummy variables interacted with *Target Advisor* and *Acquirer Advisor*, where the target (acquirer) advisor dummy equals one if the bank served as advisor to the target (acquirer) for the deal in question, zero otherwise. The coefficient on *Quarter -1* (and analogously *Quarter -2*) measures the extent to which non-advisor banks and

**Figure 3****Stock ownership by banks in target firm, ahead of merger announcement**

The sample consists of 912 mergers between 2000 and 2016, as described in Table 1. The figure shows target advisors' and acquirer advisors' stock holdings in the target firm, measured as $\frac{\text{Shares Held}}{\text{Shrs Out}}$. For each deal, we calculate an advisor-weighted average across all deals, where more weight is given to deals with more target (solid line) or acquirer (dashed line) advisors.

hedge funds increased their options holdings relative to the benchmark period of Quarters -6 through -3 , while the interaction terms capture the incremental effects for advisor banks. Our choice of two quarters allows for the possibility that institutions take informed positions at different points in time, relative to the merger announcement. As discussed earlier, the median time between the beginning of the takeover process and the public merger announcement is three to seven months.

We include a number of control variables, which are potentially correlated with options holdings. As discussed by Mayhew and Mihov (2004), demand for options trading should be influenced by the heterogeneity of investors' beliefs, and thus we include stock volume, stock volatility, and options volume. In addition, investors are likely to hold more options in companies for which there is more information disseminated (generating demand for trading) and companies with a larger investor base. Such factors suggest that market capitalization will also be positively related to the extent of options holdings. A measure of industry merger activity, defined as the percentage of firms within the same Fama-French 48-industry group that have been acquired within the past year, is included to account for the effects of merger waves (Andrade, Mitchell, and Stafford 2001). Song and Walkling (2000) conclude that firms experience positive abnormal returns when a rival firm is acquired, because of an increased probability that they will be acquired. Such effects potentially cause investors to increase their holdings of call options of firms in an industry undergoing a merger wave. A rumor dummy equals one if there were any merger-related

Table 2
Options holdings prior to merger announcement

Dependent variable	<i>Call Ratio</i>	<i>Put Ratio</i>	<i>Net Call Ratio</i>
<i>Quarter -1</i>	0.211*** (0.075)	0.004 (0.065)	0.263*** (0.067)
<i>Quarter -1*Target Advisor</i>	4.216*** (1.576)	0.831 (1.208)	3.657*** (1.387)
<i>Quarter -1*Acquirer Advisor</i>	4.775** (2.058)	1.431 (1.743)	3.381* (1.786)
<i>Quarter -2</i>	0.165*** (0.063)	-0.028 (0.050)	0.158*** (0.049)
<i>Quarter -2*Target Advisor</i>	0.759 (1.524)	1.790* (1.005)	-0.386 (1.317)
<i>Quarter -2*Acquirer Advisor</i>	3.336*** (1.209)	2.852** (1.188)	-0.181 (1.163)
<i>Merger Rumor</i>	0.708*** (0.224)	0.573*** (0.157)	0.160 (0.131)
<i>Log(Mkt Cap)</i>	0.261** (0.111)	0.028 (0.091)	0.113 (0.087)
<i>Volatility</i>	0.978 (3.379)	3.973* (2.308)	0.312 (2.618)
<i>Log(Option Volume)_T</i>	0.228*** (0.038)	0.199*** (0.027)	0.052*** (0.020)
<i>Log(Option Volume)_{T-1}</i>	-0.130*** (0.022)	-0.098*** (0.015)	-0.012 (0.014)
<i>Log(Share Volume)_T</i>	0.401*** (0.091)	0.199*** (0.067)	0.052 (0.072)
<i>Return_t</i>	-0.081 (0.081)	-0.076* (0.046)	0.028 (0.055)
<i>Return_{t-1}</i>	-0.052 (0.068)	-0.056 (0.062)	0.059 (0.042)
<i>Leverage</i>	0.602 (0.537)	0.699 (0.504)	-0.280 (0.462)
<i>Industry Merger Activity</i>	2.326* (1.199)	2.913** (1.315)	0.912 (1.392)
<i>Target Advisor</i>	1.030* (0.600)	0.227 (0.414)	0.841 (0.628)
<i>Acquirer Advisor</i>	0.908 (0.683)	0.750 (0.594)	0.457 (0.698)
<i>Lagged Dependent Variable</i>	0.727*** (0.015)	0.771*** (0.015)	0.440*** (0.018)
Number clusters	912	912	912
Fixed effects	Deal, Institution, Qtr	Deal, Institution, Qtr	Deal, Institution, Qtr
Observations	572,625	572,625	572,625
Adjusted R ²	0.525	0.567	0.172

The sample consists of 912 mergers between 2000 and 2016, as described in Table 1. The panel regression sample consists of the 912 deals $\times$ 6 quarters prior to the merger announcement $\times$ number of advisor banks and hedge funds in existence at the time of each merger. In total, there are 27 advisor banks, defined as banks that served as advisor in at least 10 sample deals and have a trading arm that participates in the options markets, and 218 hedge funds. For each deal, we identify whether each institution is advisor to the target, advisor to the acquirer, or neither. The dependent variable equals *Call Ratio* (= *Call Options* as a percentage of millions of shares outstanding) in Column 1, *Put Ratio* (= *Put Options* as a percentage of millions of shares outstanding) in Column 2, and *Net Call Ratio* (= *Calls - Puts* as a percentage of millions of shares outstanding) in Column 3. Quarter -1 equals 1 for the last quarter ending prior to the merger announcement, 0 otherwise. Quarter -2 is defined analogously, but for the prior quarter. *Target (Acquirer) Advisor* equals 1 if the institution is the target (acquirer) advisor for that deal. All other variables are defined in the Appendix. These ordinary least squares (OLS) regressions include deal, bank, and calendar quarter fixed effects. We used bootstrap procedures based on 100 iterations to estimate standard errors, which are clustered at the deal level, as shown in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

Table 3
Changes after implementation of Volcker Rule

	Dependent variable = <i>Net Call Ratio</i>		
	Sample = Advisor banks		Sample = Hedge funds
	Mergers after an advisor bank ceased proprietary trading	All mergers 2000–2016	All mergers 2000–2016
<i>Quarter –1</i>	1.888 (3.154)	5.372*** (1.300)	0.023*** (0.003)
<i>Quarter –1*Bank Ceases Proprietary Trading</i>		–5.697* (3.121)	
<i>Quarter –1*Post 2011</i>			0.027*** (0.008)
<i>Quarter –2</i>	–2.596 (2.586)	0.960 (0.839)	0.017*** (0.002)
<i>Quarter –2*Bank Ceases Proprietary Trading</i>		–4.518 (2.832)	
<i>Quarter –2*Post 2011</i>			0.012*** (0.003)
<i>Bank Ceases Proprietary Trading dummy</i>		–1.319 (2.135)	
Number clusters	395	1,145	1,145
Fixed effects	Deal, Institution, Qtr	Deal, Institution, Qtr	Deal, Institution, Qtr
Controls	Yes	Yes	Yes
Observations	4,278	17,173	631,337
Adjusted R^2	0.151	0.140	0.168

Eleven of the 27 banks in our sample ceased proprietary trading during our sample period, due to the Volcker Rule (see Table A1 for details). In Column 1, the sample is mergers in which one or more advisors had ceased proprietary trading operation $\times$ advisor(s) that had ceased proprietary trading. The dependent variable equals *Net Call Ratio*, measured as $\frac{\text{Calls} - \text{Puts}}{\text{Shrs Out}}$, where shares outstanding is expressed in millions and the ratio is expressed in percentage terms. Explanatory variables include *Quarter –1*, *Quarter –2*, and controls, as described in Table 2. Column 2 includes all advisor bank $\times$ mergers over 2000–2016. *Quarter –1* and *Quarter –2* are interacted with *Bank Ceases Proprietary Trading*, which equals 1 in periods after the advisor bank ceased proprietary trading. In Column 3, the sample consists of hedge funds over 2000–2016, and *Quarter –1* and *Quarter –2* are interacted with a dummy variable equal to 1 for years 2011 and later, as this is the median year in which banks within our sample became compliant with the Volcker Rule. Control variables (not reported) include all variables previously used in Table 2. We use bootstrap procedures based on 100 iterations to estimate standard errors, which are clustered at the deal level and shown in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

rumors concerning the firm during the quarter, as per the RavenPack data. Finally, we also control for leverage and for contemporaneous and lagged returns. Full definitions of all variables are provided in the Appendix. Deal, institution, and calendar quarter fixed effects are included in all regressions. We use bootstrap procedures based on one hundred iterations to estimate standard errors, which are clustered at the deal level.

Column 1 indicates that advisors become increasingly likely to hold calls ahead of the merger: the coefficients on *Quarter –1*Target Advisor* and *Quarter –1*Acquirer Advisor* are both significantly positive. In contrast, we find no evidence that advisors significantly increase holdings of put options in the target firm ahead of the merger, as the coefficients on these interactions are insignificant at conventional levels. Consistent with this difference, Column 3 shows that advisor banks' net call ratio significantly increases in the quarter prior to the merger announcement. In addition, the coefficients on

Table 4
Options holdings of advisor banks in target firms versus matched control firms, prior to merger announcement

Dependent variable	Call Ratio	Put Ratio	Net Call Ratio
<i>Quarter -1</i>	-0.7287 (1.3763)	-0.1701 (1.5720)	-1.0552 (1.3202)
<i>Quarter -1*Target Firm</i>	5.7693*** (2.0583)	1.7357 (1.7955)	4.0714** (1.9018)
<i>Quarter -2</i>	0.0464 (1.3824)	0.5039 (1.0401)	0.5873 (1.1409)
<i>Quarter -2*Target Firm</i>	2.1723 (2.3284)	3.1899* (1.9340)	-1.3048 (1.7558)
<i>Target Firm</i> (1.0263)	0.4989 (0.7494)	-0.1746 (0.9665)	0.8728
Number clusters	912	912	912
Fixed effects	Deal, Institution	Deal, Institution	Deal, Institution
Controls	Yes	Yes	Yes
Observations	27,315	27,315	27,315
Adjusted R ²	0.460	0.461	0.173

The sample consists of 912 mergers between 2000 and 2016, as described in Table 1. For each target firm, we identify a matched control firm based on a propensity score model. The control firm represents the non-target firm with the highest predicted probability (excluding the actual target) of being the target firm,

$$\begin{aligned} \text{Target Firm} = & \alpha + \beta_1 \log(\text{MktCap})_t + \beta_2 \text{Volatility}_{Q_{t-1}} + \beta_3 \text{Log}(\text{Options Volume})_{Q_{t-1}} + \beta_4 \text{Log}(\text{Options Volume})_{Q_{t-1}-1} \\ & + \beta_5 \text{Share Volume}_{Q_{t-1}} + \beta_6 \text{Stock Return}_{Q_{t-1}} + \beta_7 \text{Stock Return}_{Q_{t-1}-1} + \beta_8 \text{Leverage}_t \\ & + \beta_9 \text{Industry Merger Activity} + \text{Quarter FE} + \text{Fama-French 48 Industry FE} + \varepsilon \end{aligned}$$

where this regression is estimated four quarters prior to the merger announcement, and control firms are required to have publicly traded options. The regression sample for this table consists of advisor banks' options holdings (where advisors include both target and acquirer advisors) in target and matched control firms, over the six quarters prior to the merger announcement. The target dummy equals 1 for the target firms and 0 for the matched control firms. The dependent variable equals *Call Ratio* (= *Call Options* as a percentage of millions of shares outstanding) in Column 1, *Put Ratio* (= *Put Options* as a percentage of millions of shares outstanding) in Column 2, and *Net Call Ratio* (= *Calls* - *Puts*) as a percentage of millions of shares outstanding) in Column 3. All other variables are defined in Table 2 and in the Appendix. Control variables (not reported) include all variables previously used in Table 2. Regressions are OLS and include deal, bank, and calendar quarter fixed effects. We use bootstrap procedures based on 100 iterations to estimate standard errors, which are clustered at the deal level, as shown in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

*Quarter -1*Target Advisor* and *Quarter -1*Acquirer Advisor* are significantly different than those on the analogous *Quarter -2* interactions, indicating that advisors' positions increase significantly between *Quarter -2* and *Quarter -1*.

In economic terms, in the quarter immediately preceding the merger announcement target (acquirer) advisors' call ratios increase by 4.4 (5.0) percentage points, while their put ratios increase by only 0.8 (1.4) percentage points.⁸ Consistent with these differences, Column 3 indicates that target (acquirer) net call ratios increase by 3.9 (3.6) percentage points, a stark contrast to the analogous 0.3 percentage point increase for non-advisors.⁹As

⁸ For target advisors' call option holdings, economic significance is calculated as coefficient on *Quarter -1* (= 0.211) + coefficient on *Target Advisor*Quarter -1* (= 4.216). Other economic significance measures are computed analogously.

⁹ Because each regression includes a lagged dependent variable, the explanatory variables are not exactly the same across columns. This causes the coefficient on net call ratio to not exactly equal the coefficient on *Call Ratio* minus that on *Put Ratio*. Similar effects also influence regressions reported in subsequent tables.

an additional metric of economic significance, Section 4.3 provides back-of-the-envelope estimates of the dollar returns associated with these positions.

To the extent that these patterns are driven by bank proprietary trading in response to an information advantage, they should not exist after a bank ceases proprietary trading. Table 3 examines this prediction specifically. Column 1 focuses on the set of advisor bank $\times$ mergers that are omitted from our main sample due to banks' adherence to the Volcker Rule. Specifically, it includes mergers in which one or more advisors had ceased proprietary trading operations $\times$ advisor(s) that had ceased proprietary trading, 395 mergers over the 2010–2016 period. Consistent with expectations, we find no evidence within this sample of advisors' net call ratios increasing in the quarter prior to the merger announcement: the coefficient on *Quarter* -1 is insignificant at conventional levels. Column 2 similarly includes just advisor banks, but it extends the sample to deals over the entire 2000–2016 period—that is, the sample consists of the union of deals in Table 2 plus the more recent deals in Column 1 of Table 3. We define a bank-specific dummy variable, *Bank Ceases Proprietary Trading*, equal to one for all calendar quarters after a bank ceases proprietary trading for reasons related to the Volcker Rule. Results provide added evidence of a significant change in banks' behavior. Prior to the Volcker Rule, advisor banks significantly increased their call relative to put holdings prior to the merger announcement, as evidenced by the significantly positive coefficient on *Quarter* -1 in Column 2. However, these banks became significantly less likely to engage in such informed proprietary trading after their adherence to the Volcker Rule, as evidenced by the significantly negative coefficient on *Quarter* $-1 \times$ *Bank Ceases Proprietary Trading*. Finally, the sum of the coefficients on the *Quarter* -1 dummy and the interaction term is insignificantly different from zero, consistent with inferences from Column 1. These results highlight the dramatic effect that this regulatory change had on bank behavior.

The final column examines whether hedge funds filled in the gap left by advisor banks, benefiting from some information advantage that enables them to predict target firms and increase their net call ratios in these firms. Following the shutting of banks' proprietary trading departments, many of the people that had worked in these departments moved to hedge funds.¹⁰ If these people maintain their connections to banks or the clients the banks serve, they may be more likely to have access to private information. To test this, we limit the sample to hedge funds $\times$ target firms $\times$ six calendar quarters prior to the merger announcement, and we include both *Quarter* -1 and this dummy interacted with a *Post 2011* dummy (2011 is the median year in which banks in our sample closed their proprietary trading departments). We find a significantly positive coefficient on *Quarter* -1 , indicating that hedge funds were successfully predicting target

¹⁰ See, e.g., “Hotshot traders leave Wall Street,” *Wall Street Journal*, June 3, 2010.

firms throughout the sample period. However, this tendency increased markedly in the wake of the Volcker Rule, as indicated by the significantly positive coefficient on the *Quarter -1*Post 2011* interaction term. We note that the inclusion of calendar quarter dummies means that we are not merely capturing a time-series trend. As further confirmation of this, we respecify this regression using a *Post 2005* dummy instead of a *Post 2011* dummy (not tabulated). Neither the *Post 2005* dummy nor the associated interaction term is significant at conventional levels. In sum, Table 3 shows that the Volcker Rule significantly decreased banks' trading on client firm information. Interestingly, hedge funds appear to have simultaneously increased their informed trading.

Specifications in Tables 2 and 3 have the advantage of focusing on the incremental increase in advisor bank holdings, thereby controlling for any pre-merger rumors that cause all parties to alter their options holdings (that may not be fully captured by the *Merger Rumor* variable). Bank and firm fixed effects control for differences in institutions' general tendency to hold options in firms as well as differences in options holdings across firms. As an alternative way to control for bank-specific effects, Table 4 uses a specification where we compare advisor banks' options holdings in target firms versus their holdings in matched control firms, where control firms are selected using a propensity score approach, as described previously. We regress the options holdings of the advisor bank during quarter t in firm i on *Quarter -1*, *Quarter -2*, and each of these dummies interacted with Target Firm, which equals one if firm i is a target firm and zero if it is a matched control firm. Similar to Table 2, the sample includes six quarters prior to the merger announcement, and we utilize three measures of options holdings: call ratio, put ratio, and net call ratio.

Table 4 focuses on our main sample of 912 mergers, for which at least one advisor bank engages in proprietary trading, as described previously. These specifications yield conclusions broadly similar to those from Table 2: advisors are significantly more likely to hold calls versus puts in target firms, compared with matched control firms, in the quarter prior to the merger announcement. In economic terms, advisors increase their net call ratio positions by 3.0 percentage points, in the quarter immediately prior to the merger announcement, compared with the benchmark period of Quarters -6 to -3.¹¹ This is a similar magnitude to that documented in Table 2.

It is not clear why advisors would consistently be better than non-advisors in predicting these mergers if they were not somehow benefitting from extra information. Nevertheless, our findings do not "prove" insider trading in any particular case. The apparent contrast between our findings and the relatively low rate of SEC investigations into insider trading reflects both the high burden of proof in such cases and the limited investigative resources. Analogously, in an analysis of target firms' options prices prior to the merger announcement,

¹¹ The increase is represented by the sum of the coefficients on *Quarter -1* and *Quarter-1*target firm*.

Augustin, Brenner, and Subrahmanyam (2015) report that 44% of targets have excess implied volatility, but the SEC litigates only about 7% of deals.

3.2 Advisor bank trading in acquirer firms

We predict that the slightly negative average returns to acquirers will cause trading in acquirers ahead of the merger announcement to be concentrated in puts, with magnitudes that are lower than those observed for target firms. In Table 5, we conduct an analysis similar to that of Table 2, but focusing on options in the acquirer firm rather than the target firm. Panel A examines the full sample of 675 deals in which the acquirer is publicly traded and the bank has not ceased trading due to the Volcker Rule. Panel B limits the sample to the 509 deals that are more likely to have a meaningful impact on acquirer firm value, defined as deals in which relative size (defined as target market capitalization/acquirer market capitalization, three months before deal announcement) is 5% or more. Consistent with predictions, across the full sample there is little evidence of advisors changing their positions in acquirer firms in ways similar to what we observed for target firms. Moreover, after limiting the sample to deals with a relative size of 5% or more (panel B), patterns differ even more starkly from those of target firms. Both target and acquirer advisors significantly increase their holdings of puts (Column 2), and coefficients on interaction terms in the net call ratio regression (Column 3) are both negative (albeit not significant). In sum, consistent with Chan, Ge, and Lin (2015), Ordu and Schweizer (2015), and Augustin, Brenner, and Subrahmanyam (2015), we find evidence of informed trading in the options of acquirers as well.

4. Sources of Target Advisors' Information Advantage

4.1 Firm, deal, and bank characteristics

We posit that both the probability of information leakage and the probability of the asset management division purchasing target firm options in response to information obtained are greater in some cases than others. Table 6 examines four factors that potentially relate to the extent of informed options trading: the feasibility of trading options, the bank's information advantage, the bank's governance structure, and deal-specific characteristics.

Regarding the feasibility of trading options, our first proxy is *Options Volume/Stock Volume*, with higher values of this measure arguably associated with relatively lower options trading costs. To ensure that reverse causality does not affect our results, we measure this variable at the end of the prior quarter. We also employ as a proxy *High Options Institution*, under the premise that among institutions that are more active in the options market, any one position is less likely to attract attention. For each bank and hedge fund in the sample, we identify the number of publicly traded firms in which it holds options. We classify entities as high option holders if they are above the median along this

Table 5
Options holdings of advisor versus non-advisor banks in acquirer firm, prior to merger announcement

Panel A: All deals

Dependent variable	Call Ratio	Put Ratio	Net Call Ratio
<i>Quarter -1</i>	0.668 (0.806)	-0.021 (0.745)	0.622 (0.750)
<i>Quarter -1*Target Advisor</i>	0.258 (13.895)	18.545 (12.114)	-15.404 (15.442)
<i>Quarter -1*Acquirer Advisor</i>	28.781 (17.788)	9.507 (15.614)	26.407 (18.994)
Number clusters	675	675	675
Fixed effects	Deal, Institution, Qtr	Deal, Institution, Qtr	Deal, Institution, Qtr
Controls	Yes	Yes	Yes
Observations	385,729	385,729	385,729
Adjusted R^2	0.665	0.680	0.301

Panel B: Deals in which relative size $\geq 5\%$

<i>Quarter -1</i>	-0.415 (0.607)	-0.091 (0.469)	-0.380 (0.567)
<i>Quarter -1*Target Advisor</i>	1.767 (12.027)	22.938*** (8.467)	-16.886 (12.962)
<i>Quarter -1*Acquirer Advisor</i>	19.156 (17.091)	22.115* (13.176)	3.904 (17.930)
Number clusters	509	509	509
Fixed effects	Deal, Institution, Qtr	Deal, Institution, Qtr	Deal, Institution, Qtr
Controls	Yes	Yes	Yes
Observations	295,126	295,126	295,126
Adjusted R^2	0.605	0.617	0.329

The sample consists of 675 mergers between 2000 and 2016 as described in Table 1, with the further requirement that the acquirer be publicly traded. The panel regression sample consists of the 675 deals $\times$ 6 quarters prior to the merger announcement $\times$ number of advisor banks and hedge funds in existence at the time of each merger. In total, there are 27 banks, defined as banks that served as advisor in at least 10 sample deals and have a trading arm that participates in the options markets, and 218 hedge funds. Panel B limits the sample to deals in which the target market capitalization is at least 5% of the acquirer's market capitalization, measured three months prior to the deal announcement. The dependent variable equals *Call Ratio* (= *Call Options* as a percentage of millions of shares outstanding) in Column 1, *Put Ratio* (= *Put Options* as a percentage of millions of shares outstanding) in Column 2, and *Net Call Ratio* (= (*Calls* - *Puts*) as a percentage of millions of shares outstanding) in Column 3. *Quarter -1* equals 1 for the last quarter ending prior to the merger announcement, 0 otherwise. *Quarter -2* is defined analogously, but for the prior quarter. *Target (Acquirer) Advisor* equals 1 if the institution is the target (acquirer) advisor for that deal. All other variables are defined in the Appendix. Control variables analogous to those used in Table 2 but defined for the acquirer firm are included, but not tabulated. Regressions are OLS and include deal, bank, and calendar quarter fixed effects. We use bootstrap procedures based on 100 iterations to estimate standard errors, which are clustered at the deal level, as shown in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

dimension. On average, high options banks hold options in 365 publicly traded firms in a quarter, compared with 3.4 options held by low options banks.

As a proxy for the second factor, the bank's information advantage, we employ *Same Advisor Before*, defined as one if the firm employed the bank as advisor in a previous equity offering or M&A transaction within the previous three years, zero otherwise. We posit that past relationships contribute to the likelihood that the bank's asset management division has an information advantage—for example, insight into whether firm management would consider being acquired. We also employ as a proxy *Number of Advisors* in the merger, following Acharya and Johnson's (2010) finding that information leakage is positively related to the size of the financing syndicate in buyouts.

Table 6
Do advisor banks' options holdings increase more those of non-advisor banks in certain cases?

Panel A: Sample of advisor banks, proxies for likelihood of informed trades

Proxy	Dependent variable = <i>Net Call Ratio</i>								
	Feasibility of trading in options		Bank's information advantage		Bank governance			Deal-specific terms	
	<i>Options Volume / Stock Volume</i>	<i>High Options Institution</i>	<i>Same Advisor Same Before</i>	<i># Advisors</i>	<i>Poison Pill</i>	<i>CEO–Chair Duality</i>	<i>Classified Board</i>	<i>Cash Merger</i>	<i>Target Termination Fee</i>
<i>Quarter – I * Proxy</i>	43.875*** (16.712)	4.574*** (1.560)	6.112** (3.055)	0.214 (1.482)	6.131** (3.051)	5.568* (3.368)	0.884 (2.533)	–0.498 (2.236)	–0.669 (3.596)
Fixed effects	Quarter, Institution	Quarter, Deal	Quarter, Institution	Quarter, Institution	Quarter, Deal	Quarter, Deal	Quarter, Deal	Quarter, Institution	Quarter, Institution
Controls	YES	YES	YES	YES	YES	YES	YES	YES	YES
Number of clusters	912	912	912	912	832	832	832	912	912
Observations	12,447	12,447	12,447	12,447	9,033	9,045	9,045	12,447	12,447
Adjusted <i>R</i> ²	0.200	0.121	0.199	0.197	0.099	0.098	0.098	0.197	0.198

Table 6
(Continued)

Panel B: Sample of advisor and non-advisors, proxies for likelihood of informed trades

Proxy	Dependent variable = <i>Net Call Ratio</i>								
	Feasibility of trading in options	<i>High Options Institution</i>	Bank's information advantage	# Advisors	Bank gover- nance		<i>Classified Board</i>	Deal-specific terms	
	<i>Options Volume / Stock Volume</i>		<i>Same Advisor Same Before</i>		<i>Poison Pill</i>	<i>CEO–Chair Duality</i>		<i>Cash Merger</i>	<i>Target Termination Fee</i>
<i>Quarter – 1 * Proxy * Advisor</i>	56.633*** (15.473)	6.189*** (1.300)	5.714 (3.633)	0.731 (1.50)	5.457* (3.092)	4.460 (3.603)	0.590 (2.648)	–1.17 (2.21)	–1.82 (3.76)
Fixed effects	Quarter, Institution	Quarter, Deal	Quarter, Institution	Quarter, Institution	Quarter, Deal	Quarter, Deal	Quarter, Deal	Quarter, Institution	Quarter, Institution
Controls	YES	YES	YES	YES	YES	YES	YES	YES	YES
Number of clusters	912	912	912	912	832	832	832	912	912
Observations	570,267	572,625	572,625	572,625	49,505	50,250	50,250	572,625	572,625
Adjusted <i>R</i> ²	0.175	0.170	0.174	0.174	0.146	0.147	0.146	0.174	0.174

The sample consists of 912 mergers between 2000 and 2016, as described in Table 1. The panel regression sample in panel A consists of each advisor bank's holdings (including both target and acquirer advisors) in each target firm, in each of six quarters prior to the merger announcement. The sample in panel B also includes the holdings of non-advisors, where non-advisors include both banks and hedge funds. In panel A, the independent variable of interest is *Quarter – 1 * Proxy for likelihood of informed trades * Advisor*. We examine four sets of proxies. Proxies for the feasibility of trading options include lagged $\frac{\text{Options Volume}}{\text{Stock Volume}}$ and a *High Options Institution* dummy, equal to 1 if the institution holds an above-median number of options. Proxies for the institution's information advantage include *Same Advisor Before*, equal to 1 if the bank served as advisor to the target firm in either an equity offering or merger within the past three years and 0 otherwise, and *# Advisors*, equal to the number of advisors in the deal. Proxies for bank governance include dummy variables denoting whether the bank has a poison pill, whether the CEO is also chairman of the board (CEO-chair duality), and whether the bank has a classified board. Proxies for deal terms include *Cash Merger*, equal to 1 for cash mergers and 0 otherwise, and *Target Termination Fee*, equal to 1 if the target firm has to pay a fee to terminate the deal and 0 otherwise. The dependent variable equals *Net Call Ratio*, defined as $(\text{Calls} - \text{Puts})$ as a percentage of millions of shares outstanding. All other variables are defined in Table 2 and in the Appendix. Control variables used in Table 2 are included, but not tabulated. Regressions are OLS, have robust standard errors clustered at the deal level, and fixed effects as denoted at the bottom of each column. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

The third factor, bank governance structure, is based on the idea that in banks with stricter monitoring, there will be less illicit activity. Based on prior literature, we employ several proxies for the bank's corporate governance, including whether the bank has a *Poison Pill*, a *Classified Board*, and *CEO Chair Duality*, defined as the CEO also being chairman of the board (see, e.g., Bebchuk, Coates IV, and Subramanian 2002 and Adams, Hermalin, and Weisbach 2010).

Finally, the fourth factor, deal-specific terms, enables us to ascertain whether advisors' information advantages relate not only to the probability of an upcoming merger but also to the specific terms of that merger. Our two proxies include *Cash Deal* and *Target Termination Fee*. The benefits of taking positions in deals with these features are greater, as cash mergers tend to have higher premia and deals with termination fees are less likely to be withdrawn.

Panels A and B of Table 6 examine the influence of these factors in two ways. In panel A, we restrict the sample to firm $\times$ bank pairs in which the bank served as advisor on either the target or acquirer side, and for each of these target firm $\times$ bank pairs we include six quarters prior to merger announcement. Panel B broadens the analysis to all target firm $\times$ institution pairs, where institutions include advisor and non-advisor banks and hedge funds, thus comparing changes in options positions across advisors versus non-advisors. In sum, panel A focuses on two-way interactions (*Quarter -1 * Likelihood of informed trading proxy*), whereas panel B focuses on three-way interactions (*Quarter -1 * Likelihood of informed trading proxy * Advisor*). For conciseness, we define a single advisor dummy, equal to one if the bank served as advisor to either the target or acquirer. In specifications where the proxies of interest are at the firm-level (e.g., *Same Advisor Before*, *Options Volume/Stock Volume*), we omit deal fixed effects but include bank and calendar quarter fixed effects. Analogously, in cases where proxies are at the investor level (e.g., high options institutions), we omit institution fixed effects but include deal and calendar quarter fixed effects. The dependent variable in all specifications is *Net Call Ratio*, and robust standard errors are clustered at the deal level.

The first set of columns provides strong evidence that the likelihood of the advisor bank engaging in informed trading relates to the feasibility of trading in options. The significantly positive coefficients on the interaction terms in Columns 1 and 2 are consistent with advisor banks' net call ratios increasing significantly more when the target firm's options are more liquid and when the bank participates more regularly in the options market. In economic terms, advisors' net call ratios in Quarter -1 increase by only 0.8 percentage points for firms at the 25th percentile of options volume/stock volume, compared with 4.3 percentage points for firms at the 75th percentile. Similarly, the analogous increases are 0.9 percentage points for low options institutions, compared with 5.5 percentage points for high options institutions. Moreover, panel B shows that this effect is significantly greater for advisor banks than for non-advisor banks, where the coefficients on the analogous triple interactions are both significant at the 1% level.

Panel A of Table 6 also suggests the extent of informed trading is related to the bank's information advantage in the firm. Specifically, advisor banks are significantly more likely to increase their net call ratios in firms for which they have previously served as advisor. This is consistent with findings of Cohen, Frazzini, and Malloy (2010) and Kuhnen (2009) that past connections facilitate information transfer. However, in panel B regressions where the sample includes both advisor and non-advisor institutions, we find no evidence that these factors provide a disproportionate advantage for advisor banks. The coefficients on the triple interaction terms are insignificant at conventional levels. In aggregate, these findings suggest that a prior advisory relationship contributes to an information advantage, but this information channel is not unique to advisor banks. Contrary to Acharya and Johnson (2010), we find no evidence that the number of advisors plays a significant role.

The third set of columns provides some evidence that the bank's governance structure is also a relevant factor. The presence of a poison pill and CEO-chair duality are both associated with weaker governance environments, and both are associated with a greater tendency to increase net call ratios in the quarter ahead of the merger announcement. However, we find no evidence that *Classified Board* has a significant effect.¹² Also, panel B provides little evidence of corporate governance playing any role in explaining the behavior of advisors relative to non-advisor banks. In sum, evidence regarding the influence of the bank's governance structure is mixed.

The last set of columns provides no evidence that advisors are significantly more likely to take informed positions in certain types of deals. In aggregate, our evidence suggests that while the proprietary trading division obtains sufficient information to assess an increased probability of deal, they are not privy to deal-specific terms.

4.2 Endogeneity concerns

The economically and statistically significant differences between advisors' and non-advisors' options positions leading up to merger announcements are consistent with banks obtaining an information advantage in client firms as a result of their advisory role. A possible alternative explanation for our results is that other factors—for example, a prior relationship between the bank and the firm—cause both the bank to be selected as advisor and the asset management division to have insight into the upcoming merger.

If the true channel of information transfer is through the investment banking division, then banks that are similar on other dimensions, in the sense that they had a high ex ante probability of being selected as advisor, should not have a similar information advantage. To investigate this possibility, we develop

¹² We are only able to obtain bank-governance measures for 12 distinct banks within our sample. In addition, poison pill, classified board, and CEO-chair duality are positively correlated at approximately the 20% level. All of these factors suggest that the governance results should be interpreted with caution.

a model predicting the probability that a bank will be selected as advisor to the target firm. We label the bank with the highest ex ante probability of being selected (excluding the bank that was actually selected) the pseudo target advisor. Our focus on target advisors is driven by two factors: (i) the existence of non-public acquirers limits our ability to develop a model predicting acquirer advisors, and (ii) inclusion of pseudo target advisors and actual acquirer advisors within a single regression provides an informative contrast.

Our regression model to estimate pseudo-advisors employs a panel data set consisting of total number of deals $\times$ total potential advisors, and the dependent variable equals one if the advisor was an actual target advisor on the deal, zero otherwise. Independent variables include measures for the likelihood that a particular bank will be chosen by a particular target firm. First, to capture the tendency of larger firms to employ top investment banks, we put target firms into quintiles based on market capitalization and advisor banks into quintiles based on league table rankings.¹³ We predict that the absolute value of the difference between these quintiles will be negatively related to the probability of a target firm—bank match. Our second measure captures the fact that banks with a larger M&A department, who advise in more deals, are more likely to be selected as advisor. The third and fourth measures represent dummy variables indicating whether the bank served as advisor to the target on any merger, seasoned equity offering (SEO), or initial public offering (IPO) within the past three years. The importance of relationships suggests that targets will be more likely to select a bank they have previously employed. Results of the regression are strongly consistent with predictions:

Logit(Target Advisor)

$$= -380 + -0.28^* / \text{Target Mcap Quintile} - \text{Bank League Table Quintile} / \\ + 0.002^* \text{Total Bank M\&A Activity} + 2.48^* \text{Bank Advised Target Previously} + e$$

The coefficients on each of the independent variables are significant in the predicted directions, with p -values less than 0.0001 in all cases.¹⁴

We estimate regressions similar to those in Table 2, with the exception that the pseudo target advisor is substituted for the actual target advisor. As shown in Table 7, results contrast markedly with those in Table 2. First, none of the coefficients on the pseudo target advisor interaction terms are significant in the predicted direction. Second, the magnitude of the coefficient on *Quarter -1*Pseudo Target Advisor* in the net call ratio regression (Column 3) decreases by close to 50%, compared with the *Quarter -1*Target Advisor* in Table 2.

¹³ League table rankings are obtained from Bloomberg.

¹⁴ The logistic regression fits the data well in the sense that an advisor ranked by the model as the most likely advisor was the actual advisor approximately 25% of the time. Conversely, there is only one case in which an advisor ranked below 20th in likelihood is selected as the actual advisor.

Table 7
What is the source of advisor banks' information?

	Call ratio	Put ratio	Net call ratio
<i>Quarter -1*Pseudo Target Advisor</i>	2.638 (1.841)	0.201 (1.424)	2.245 (1.713)
<i>Quarter -1*Actual Acquirer Advisor</i>	4.747** (1.964)	1.423 (1.583)	3.355** (1.713)
<i>Quarter -2*Pseudo Target Advisor</i>	-1.512 (1.336)	0.671 (1.487)	-2.583* (1.341)
<i>Quarter -2*Actual Acquirer Advisor</i>	3.310** (1.364)	2.836** (1.360)	-0.203 (1.291)
<i>Pseudo Target Advisor</i>	-0.829 (0.740)	-0.372 (0.577)	-0.174 (0.718)
<i>Actual Acquirer Advisor</i>	0.558 (0.682)	0.626 (0.533)	0.232 (0.642)
<i>Quarter -1</i>	0.238*** (0.072)	0.015 (0.065)	0.283*** (0.065)
<i>Quarter -2</i>	0.188*** (0.065)	-0.013 (0.051)	0.174*** (0.050)
Lagged dependent variable	0.727*** (0.015)	0.771*** (0.017)	0.440*** (0.018)
Fixed effects	Deal, Institution, Quarter	Deal, Institution, Quarter	Deal, Institution, Quarter
Controls	Yes	Yes	Yes
Number of clusters	912	912	912
Observations	572,625	572,625	572,625
Adjusted R ²	0.524	0.566	0.171

The sample consists of 912 mergers between 2000 and 2016, as described in Table 1. The regression sample consists of the 912 deals $\times$ 6 quarters prior to the merger announcement $\times$ number of advisor banks and hedge funds participating in options markets. For each target advisor in each deal, we estimate a pseudo target advisor based on a propensity score model. The pseudo target advisor represents the non-advisor bank with the highest predicted probability (excluding the actual advisor) of being the advisor:

$$\begin{aligned} \text{Target Advisor} = & \alpha + \beta_1 / \text{Target Mkt Cap Quintile} - \text{Bank League Table Quintile} / + \beta_2 \text{Total Bank M\&A Activity} \\ & + \beta_3 \text{Bank Advised Target Previously} + \varepsilon \end{aligned}$$

The dependent variable equals *Call Ratio* (= *Call Options* as a percentage of millions of shares outstanding) in Column 1, *Put Ratio* (= *Put Options* as a percentage of millions of shares outstanding) in Column 2, and *Net Call Ratio* (= (*Calls* - *Puts*) as a percentage of millions of shares outstanding) in Column 3. *Pseudo Target (Actual Acquirer) Advisor* equals 1 if the bank is the pseudo target (actual acquirer) advisor for that deal. All other variables are defined in Table 2 and in the Appendix. Control variables previously used in Table 2 are included, but not tabulated. Regressions are OLS and include deal, bank, and calendar quarter fixed effects. We use bootstrap procedures based on 100 iterations to estimate standard errors, which are clustered at the deal level, as shown in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

Third, the coefficient on *Quarter -1*Actual Acquirer Advisor* continues to be significantly positive, with a magnitude similar to the analogous coefficient in Table 2. Based on these results, we conclude that being a “likely” advisor is not sufficient to gain the knowledge necessary to profit from an impending deal. Rather, it is the actual advising relationships that give banks access to private, valuable, and tradable information.

Using simulations, we also examine the extent to which similar trading patterns exist around pseudo-announcement dates. As described in Online Appendix Table A4, this analysis demonstrates that advisor banks’ tendency to significantly increase their holdings of net call ratio in the target firm is unique to the period leading up to the merger announcement date.

4.3 Profitability estimates

The strategy of increasing the net call ratio in the target firm in advance of merger announcements will clearly be profitable. To provide back-of-the-envelope estimates regarding the magnitude of these profits, we obtain information on all call options and all put options that were outstanding as of the last day of the quarter preceding the merger announcement for each of our sample firms. We implement a strategy of purchasing each call and each put at the closing price on this date, and conditional on it being in-the-money and exercising it on one of the five days closest to expiration (unless the option expires after merger completion, in which case it is exercised on one of the five days preceding completion). We calculate the profit on each call and on each put, and we weight these profits by banks' average ratio of calls to puts at the end of the quarter preceding the merger. This strategy produces a monthly return of 11.9%. These are likely conservative estimates of options profits. Profits would be substantially higher if one focused on call options that were deep out of the money prior to the merger announcement, and Augustin, Brenner, and Subrahmanyam (2015) find that option trading in the weeks preceding the merger announcement is concentrated in out-of-the money calls.

In comparison, the monthly return if one instead purchases target stock at the end of the calendar quarter prior to the merger announcement equals 5.0%, where stock returns are calculated following the approach of Bodnaruk, Massa, and Siminov (2009) and Mitchell and Pulvino (2001). In sum, the return to an informed investor from trading in options is approximately double that from trading in stock.

5. Stock Holdings of Advisor Banks, in Targets

The substantial leverage available to informed investors through the options market suggests that informed trading should be concentrated in options rather than stock ahead of information intensive events such as mergers. Figure 3 and Table 8 investigate this issue. In addition, we investigate the source of the contradictory findings regarding this issue within prior literature.

Looking first at Figure 3, we observe no evidence of advisors increasing their stock holdings in the quarter immediately prior to the merger announcement. To examine these patterns more rigorously, Table 8 shows five regressions. Column 1 focuses on whether advisor banks' holdings in target firms are particularly high prior to the merger announcement, relative to similar firms. The sample includes advisor banks' holdings of target firms and matched control firms, in each of the two quarters prior to the merger announcement. Each target firm is matched to a control firm using the propensity score approach described earlier. The stock ownership ratio (shares held over shares outstanding, as defined previously) is regressed on a target firm dummy (equal to one for the target firms and zero for matched control firms), control variables used in prior regressions, and deal fixed effects. We find a significantly positive coefficient on the target

Table 8
Tendency of investment banks to hold stock in target firm

	Dependent variable = <i>Stock Ownership Ratio</i>				
	Holdings by advisor banks, in target and matched control firms			Holdings in target firms	
	Quarters -1 and -2	Quarters -1 and -2	Quarters -1 to -6	By advisor banks Quarters -1 to -6	By all banks and hedge funds Quarters -1 to -6
<i>Target Firm</i>	0.0796** (0.0377)	0.0154 (0.0151)	0.0175* (0.0095)		
<i>Quarter -1*Target Firm</i>			-0.0009 (0.0201)		
<i>Quarter -1</i>			0.0191 (0.0190)	0.008 (0.035)	0.005*** (0.001)
<i>Quarter -2</i>				-0.011 (0.028)	0.003*** (0.001)
<i>Quarter -1*Advisor</i>					-0.006 (0.019)
<i>Quarter -2*Advisor</i>					-0.028* (0.017)
<i>Advisor</i>					0.010 (0.008)
Lagged dependent variable		0.0796*** (0.0033)	0.0805*** (0.0027)	0.074*** (0.002)	0.084*** (0.001)
Fixed effects	Deal	Deal, Qtr Institution	Deal, Qtr Institution	Deal, Qtr Institution	Deal, Qtr Institution
Controls	Yes	Yes	Yes	Yes	Yes
Number of clusters	912	912	912	912	912
Observations	10,402	10,258	30,394	12,447	572,625
Adjusted <i>R</i> ²	0.002	0.686	0.668	0.609	0.790

The dependent variable equals *Stock Ownership Ratio*, defined as shares held in the firm as a fraction of shares outstanding, expressed in percentage terms. Columns 1 and 2 (Column 3) include two quarters (six quarters) prior to merger announcement, and the sample represents advisors' holdings in target firms and matched control firms (as described in Table 4). *Target Firm* equals 1 for the actual targets and 0 for matched control firms. Columns 4 and 5 are based on a panel data set of 912 mergers * six pre-merger announcement quarters * institutional traders, where institutional traders include advisor banks in Column 4 and advisor and non-advisor banks and hedge funds that participate in options markets in Column 5. All other variables are defined in Table 2 and in the Appendix. Control variables (not reported) include all variables previously used in Table 2. Regressions are OLS and include deal fixed effects, and Columns 2–5 also include bank and calendar quarter fixed effects. Robust standard errors, clustered at the deal level, are shown in parentheses. *, **, and *** denote significance at the 10, 5, and 1% levels.

firm dummy, suggesting that advisor banks hold significantly more stock in their client firms, compared with matched control firms. Bodnaruk, Massa, and Siminov (2009) document a similar positive relation.¹⁵

Subsequent columns investigate in more depth the source of these higher holdings. If they are driven by a greater propensity to purchase equity upon learning of a pending merger, they should be robust to the inclusion of lagged stock holdings and additional fixed effects—for example, bank fixed effects that capture the tendency of some banks to hold more options in all firms. Also, advisors' holdings in target firms should increase in the quarters leading up to the merger announcement. Columns 2 and 3 examine these propositions

¹⁵ Specifically, Bodnaruk, Massa, and Siminov (2009) regress *Target Firm Dummy* = β * *Advisor Bank Stock Holdings* + *X* + *e* with deal fixed effects on a sample of target firms and matched control firms, and β is significantly positive.

directly. However, after including additional fixed effects and lagged share holdings (Column 2), the coefficient on the target firm dummy decreases substantially and it becomes insignificant. Column 3 extends the sample to six quarters prior to the merger announcement, and $Quarter -1 * Target Firm$ captures the extent to which advisor banks increase their holdings by a greater amount in the target firm relative to matched control firms, in the quarter immediately prior to the merger announcement. The coefficient on this variable is negative. In sum, we find no support for advisors making informed trades in target firm stock.¹⁶

Columns 4 and 5 examine the same question from a different perspective, with specifications more similar to our main options regressions, as reported in Table 2. In Column 4, the sample consists of advisor banks' stock holdings in the target firm over the six quarters prior to the merger announcement, and $Quarter -1$ captures any change in advisors' holdings in the target firm. Column 5 extends the sample to advisor and non-advisor banks and hedge funds, and $Quarter -1 * Advisor$ captures any changes in advisors' holdings relative to non-advisors' holdings. Neither specification provides any support for advisor banks increasing their holdings of target firm stock in advance of the merger announcement, as evidenced by the insignificant coefficients on both these variables.

In aggregate, results in this table are consistent with the findings of Griffin, Shu, and Topaloglu (2012), but inconsistent with the conclusions of Bodnaruk, Massa, and Siminov (2009). We have investigated in depth the extent to which either time period or test specification contributes to the different conclusions. First, we find some evidence that patterns have changed over time. Consistent with Bodnaruk, Massa, and Siminov (2009) reported results, we find that over the 1984–2003 period advisors' holdings of client firm stock increase in the quarter prior to the merger announcement. However, as shown in Figure 3, we do not observe similar trends over our 2000–2016 sample period.¹⁷

Second, we find strong evidence that different test specifications contribute to the contrasting conclusions. Bodnaruk, Massa, and Siminov (2009) estimates show that advisor banks hold significantly more stock in target firms than matched control firms prior to merger announcement (similar to our finding in Column 1 of Table 8). However, Table 8 shows that during our sample period, conclusions from such estimations are sensitive to regression specification. Online Appendix Table A5 shows that conclusions are similarly sensitive to

¹⁶ We have also examined whether there is evidence of advisors increasing their holdings of target firm stock among the subsample of target firms without tradeable options. However, we do not find strong evidence in support of this possibility.

¹⁷ As shown in Online Appendix Table A2, during the earlier 1984–2003 sample period the percentage of target advisors owning stock increased by 5.2 percentage points (36.7% to 41.9%) between the benchmark period (i.e., Quarters –6 through –3) and Quarter –1, compared with an increase of 2.3 percentage points among non-advisor banks. In contrast, during our sample period the percentage of target advisors owning stock increased by a lower amount than non-advisors. Similar shifts are seen in the stock ownership ratio and among acquirer advisors.

regression specification over their 1984–2003 sample period. We consistently find that the level of advisors' holdings is greater in target firms than control firms, but we find no evidence that advisors increase their holdings as the merger approaches, as one would expect if they were engaging in informed trading in the target firm stock.¹⁸ In sum, while we find some evidence of a time period effect, the primary determinant of the difference in results is test specification.

6. Relation Between Advisors' Options Holdings and Merger Returns

In this section, we examine whether advisors' options holdings are related to target firm stock returns. Cao, Chen, and Griffin (2005) find that ahead of information-intensive events, price discovery occurs first in options markets and is subsequently observed in stock prices. To the extent that advisors' options positions are informed, this suggests that the options positions will predict stock price run-ups ahead of the merger. To test this conjecture, we measure target firm run-ups as the target firm return between the end of the last quarter prior to the merger announcement and two days prior to the merger announcement, minus the return on the market index over the same period. As shown in Table 9, we regress the target firm run-ups on the change in banks' holdings at the end of the quarter immediately preceding the merger announcement. Similar to prior tables, Table 9 focuses on advisors' options holdings in Column 1, and both advisors' and non-advisors' holdings in Column 2.

This analysis is complicated by sample selection issues: banks are more likely to hold options in firms with certain characteristics, and similar firm characteristics are likely also correlated with merger returns. For example, small firms have significantly larger pre-merger run-ups, but significantly lower options volume (see, e.g., Mayhew and Mihov 2004) and significantly lower options holdings by institutions within our sample.¹⁹ Because we cannot completely control for such factors, we attempt to mitigate the problem by limiting the sample to bank-firm pairs in which the bank has positive options holdings in the firm in the quarter prior to the merger announcement. This limits the sample to 502 mergers. We then examine whether the relative holdings of calls versus puts is related to merger returns.

Consistent with predictions, results suggest that the change in the net call ratio of the advisor are significantly related to merger run-ups. Looking at Column 1, a one-standard-deviation increase in $\Delta Net\ Call\ Ratio$ is associated with a 2.6%

¹⁸ Results are robust across various subsamples, e.g., deals in which target firms are small versus large, and deals in which acquirers are publicly traded. We have also separately examined target versus acquirer advisors. (Results not tabulated.)

¹⁹ Splitting target firms by median market capitalization, small firms' average run-up is approximately twice as large (7.1% vs. 3.6%), but options volume is approximately 1/7 the magnitude of large firms (40,821 vs. 278,952). Consistent with this substantial difference in options volume, advisor banks hold options in 12.6% of small firms compared with 37.3% of large firms, and the associated average net call ratio equals 1.9% vs. 15.1% in the quarter prior to the merger announcement.

Table 9
Are target option holdings related to target firm run-up?

Dependent variable = *Target firm abnormal run-up*

	Sample	Advisor banks	Advisors and non-advisors
$\Delta \text{Net Call Ratio} * \text{Advisor}$			0.007** (0.004)
$\Delta \text{Net Call Ratio}$		0.010** (0.004)	0.001 (0.001)
$\Delta \text{Stock Ownership Ratio} * \text{Advisor}$			-0.068 (0.086)
$\Delta \text{Stock Ownership Ratio}$		-0.050 (0.084)	0.042 (0.054)
<i>Advisor</i>			-0.439 (1.314)
$\text{Log}(\text{Mkt Cap})$		-0.952 (1.468)	-1.452** (0.603)
<i>Volatility</i>		372.680*** (119.413)	270.585*** (45.665)
$\text{Log}(\text{Option Volume})_t$		-1.885 (1.610)	0.942 (0.723)
$\text{Log}(\text{Option Volume})_{t-1}$		-0.611 (1.187)	-2.134*** (0.627)
$\text{Log}(\text{Share Volume})$		0.811 (1.860)	-1.026 (0.707)
Return_t		-9.232 (5.840)	-15.065*** (3.480)
Return_{t-1}		-11.253* (5.845)	-2.903 (2.679)
<i>Leverage</i>		-4.547 (7.259)	-8.173*** (2.706)
<i>Industry Merger Activity</i>		-156.352*** (47.123)	-149.159*** (18.271)
<i>Merger Rumor</i>		-5.582* (3.186)	0.873 (1.112)
<i>Competing Bid Dummy</i>		9.992*** (3.755)	0.535 (1.672)
<i>Number Days QuarterEnd to Annt Date</i>		3.695** (1.535)	2.583*** (0.566)
Fixed effects		Year, Industry, Institution	Year, Industry, Institution
Observations		502	3372
Adjusted R^2		0.307	0.363

The sample consists of 502 mergers, as previously defined in Table 1 with the additional requirement that the advisor bank have nonzero options holdings in the quarter prior to the merger announcement. The dependent variable equals the target firm abnormal return run-up, defined as the target firm return from the end of the quarter immediately preceding the merger announcement through Day -2 relative to the merger announcement, net of the value-weighted market return over that same period. The target abnormal return is regressed on the change in the *Net Call Ratio*, the change in the *Stock Ownership Ratio*, and each of these measures interacted with *Advisor*, which equals 1 if the bank was advisor to the target or acquirer firm and 0 otherwise. These variables as well as all control variables are further defined in Table 2 and in the Appendix. All independent variables are measured at the last quarter end prior to the merger announcement with the exception of *Merger Rumor* which is measured during the quarter. In Column 1, the sample is restricted to advisor banks, and in Column 2 the sample also includes non-advisor banks and hedge funds. Industry and year fixed effects are included, and robust standard errors are reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

higher run-up, which compared with the mean run-up of 4.9% represents an approximately 53% increase. Column 2 shows that this effect is significantly greater for advisor banks, as evidenced by the significantly positive coefficient on $\Delta \text{Net Call Ratio} * \text{Advisor Dummy}$. In sum, the positive relation between

advisors' options positions and subsequent stock price run-ups is consistent with information diffusing from the options market to the stock market.

Additional analysis provides no support for the conjecture that the trading department has sufficiently detailed information on deal terms to enable them to forecast merger-related returns. Regressions of announcement returns on advisors' options holdings (not tabulated) show no significant relation. We note that the prior finding shown in Table 6 that advisors are not significantly more likely to trade in advance of cash offers, in spite of the fact that cash offers have higher average premiums, is similarly consistent with advisors' trading arms having insight on the probability of a merger, but not detailed information on deal specifics.

Informed trading by advisor banks prior to the merger announcement is potentially problematic for the companies involved in the merger. Companies generally have incentives to keep the deal quiet as a way to avoid stock price run-ups that may adversely affect the terms of an ultimate merger. Interestingly, our findings are consistent with company statements that they sometimes avoid using an advisor due to concerns regarding information leakage.

7. Discussion

7.1 Robustness

Additional analyses examine the robustness of our conclusions. First, we consider alternative benchmark samples. Our main tests are based on samples of advisor and non-advisor banks and hedge funds. Table 10 replicates the Table 2 specifications using three alternative samples. Panel A uses a broader sample of all 13F filers. Panel B employs a sample of advisor banks benchmarked against entities that never serve as advisors—that is, hedge funds. Finally, panel C restricts the sample to banks that serve as advisors in some deals, meaning we benchmark advisor banks against banks with an advisory function but which don't serve as advisor for the deal in question.

For conciseness, Table 10 reports only the main coefficients of interest from these regressions. Using each of these alternative samples, results are similar to those reported in Table 2: *Quarter -1 * Target Advisor* and *Quarter -1 * Acquirer Advisor* are significantly positively related to both the call ratio and the net call ratio.

Second, we consider an alternative measure of options holdings. Our main measure, net call ratio, has the advantage of capturing variation in the extent of banks' holdings. However, a lack of data on exercise prices and expiration dates prevents us from determining whether some players' options are more in-the-money or out-of-the-money. To further ensure the robustness of our conclusions we also estimate regressions using a simpler *Call Dummy—Put Dummy* measure, where *Call (Put) Dummy* equals one if the bank held any call (put) options in the designated firm-quarter, zero otherwise. Results are qualitatively similar using this alternative measure (not tabulated).

Table 10
Robustness

Dependent variable	Call Ratio	Put Ratio	Net Call Ratio
Panel A: Advisor banks benchmarked against non-advisor banks, hedge funds, and all other 13F filers			
<i>Quarter -1</i>	0.156*** (0.060)	0.063 (0.046)	0.200*** (0.050)
<i>Quarter -1*Target Advisor</i>	4.790** (1.869)	0.887 (1.267)	3.982** (1.606)
<i>Quarter -1*Acquirer Advisor</i>	6.163*** (2.247)	1.523 (1.595)	4.335** (1.915)
Fixed effects	Deal, Quarter, Institution	Deal, Quarter, Institution	Deal, Quarter, Institution
Number of clusters	912	912	912
Observations	905,854	905,854	905,854
Adjusted R^2	0.512	0.554	0.173
Panel B: Advisor banks benchmarked against hedge funds			
<i>Quarter -1</i>	0.077** (0.037)	0.000 (0.028)	0.080** (0.036)
<i>Quarter -1*Target Advisor</i>	4.322** (1.719)	0.858 (1.205)	3.761** (1.469)
<i>Quarter -1*Acquirer Advisor</i>	4.696** (1.954)	1.336 (1.504)	3.443** (1.707)
Fixed effects	Deal, Quarter, Institution	Deal, Quarter, Institution	Deal, Quarter, Institution
Number of clusters	912	912	912
Observations	487,134	487,134	487,134
Adjusted R^2	0.589	0.633	0.200
Panel C: Advisor banks benchmarked against non-advisor banks (excluding hedge funds)			
<i>Quarter -1</i>	1.047** (0.423)	-0.072 (0.360)	1.452*** (0.409)
<i>Quarter -1*Target Advisor</i>	3.522** (1.670)	0.421 (1.177)	3.217** (1.453)
<i>Quarter -1*Acquirer Advisor</i>	3.953** (1.942)	0.965 (1.469)	2.836* (1.713)
Fixed effects	Deal, Quarter, Institution	Deal, Quarter, Institution	Deal, Quarter, Institution
Number of clusters	912	912	912
Observations	97,938	97,938	97,938
Adjusted R^2	0.485	0.527	0.163

The sample consists of 912 mergers between 2000 and 2016, as described in Table 1, and these regressions follow the format of those shown in Table 2 (only main coefficients of interest are tabulated) with the exception that they are based on different samples of institutions. In each panel, the regression sample consists of the 912 deals $\times$ 6 quarters prior to the merger announcement $\times$ institutions. In panel A, institutions include advisor and non-advisor banks, hedge funds, and all other 13F filers. In panel B, institutions include only advisor banks and hedge funds. In panel C, institutions include only advisor and non-advisor banks. All variables are defined in Table 2 and in the Appendix. Regressions are OLS and include deal, bank, and calendar quarter fixed effects. Robust standard errors, clustered at the deal level, are shown in parentheses, and *, **, and *** denote significance at the 10%, 5%, and 1% levels.

7.2 Sensitivity of conclusions to non-observable dimensions of options trading

As discussed previously, 13F filings provide the number of call and put options but information on the respective exercise prices or exercise dates and information on options written are both unavailable. We consider in this section the extent to which this data limitation raises concerns about our conclusion that advisor banks engage in informed trading.

We first consider the possibility that the observed increases in advisor banks' net call ratios do not reflect informed bets on future price movements of the target firm. While we observe increases in advisor banks' call option holdings, these banks could simultaneously write increased numbers of call options. Because many advisors serve as market makers, this scenario may be likely, for example, if there was added demand by other investors to purchase call options in the period leading up to the merger announcement. A lack of data prevents us from completely ruling out this possibility. However, this concern is at least mitigated by the fact that relations reverse after banks become compliant with the Volcker Rule, suggesting a change in behavior that coincides with the end of proprietary trading. Because only delta-neutral market-making activities are preserved by the Volcker Rule, the change in banks' long positions more likely reflects a change in informed trading.

Second, we consider the possibility that institutions pursue other strategies to profit from information about an impending deal announcement. An informed trader can utilize options to create a synthetic stock position, or she can use alternative options positions (e.g., short puts and/or a combination of short and long calls) to achieve similar cash flow rights as the long call.²⁰ If non-advisors disproportionately utilize such strategies, then our conclusion that advisors engage in significantly more informed trading will be inaccurate. Since we cannot observe short positions on put options, we cannot rule out this possibility entirely. Nevertheless, several factors again at least partially mitigate this concern. First, Cremers, Fodor, and Weinbaum (2017) show theoretically and empirically that long call positions are preferable to short put positions ahead of unscheduled news releases, such as mergers. Second, in an analysis of 13F filings, Collins-DuFresne, Fos, and Muravyev (2017) find that informed traders take long call positions in the vast majority (83%) of cases. One would have to write quite a large number of put contracts to achieve a similar exposure to that available through call options, and collateral requirements would be substantial.²¹ Finally, Augustin, Brenner, and Subrahmanyam (2015) explicitly consider a plethora of synthetic strategies and conclude that nearly all of them either include long call positions or would be substantially riskier or more expensive than the comparative long call position. In sum, to the extent that these papers' findings are generalizable to our sample, they suggest that our focus on long positions should capture the majority of informed trading.

²⁰ This synthetic stock strategy has the added benefit of capturing the cash flow component of the stock, without having to pay for the control rights associated with direct ownership, which could be expensive at a time when change in control is probable (Kalay, Karakas, and Pant 2014). The internet appendix of Augustin, Brenner, and Subrahmanyam (2015) provides a detailed description of various option strategies.

²¹ The margin requirement for shorting puts equals 20% of the underlying security value reduced by the amount by which the option is out-of-the-money, but there is a minimum requirement of 10% of the put strike price. Hitzemann et al. (2017) report that the average (median) margin requirement across a broad set of call and put options equals 1.6 (4.0) times the option price. See also Chicago Board Options Exchange (2000).

7.3 Advisor banks versus hedge funds

Our results suggesting that the asset management side of advisor banks has an advantage in deducing the probability of upcoming mergers contrast with the results of Solomon and Soltes (2015), who conclude that hedge funds are better able than other investors to piece together information—at least some of which is not widely publicly available—to make informed trading decisions. We conjecture that the different settings of the two papers contribute to the difference in findings. Our focus on a setting in which at least some divisions of advisor banks have a clear information advantage regarding one very information-intensive event increases the probability of finding informed trading by advisor banks. In contrast, Solomon and Soltes (2015) focus on private investor meetings during regular times, when the potentially greater sophistication of hedge funds may represent a larger advantage. Moreover, they compare hedge funds to all banks, the majority of which would presumably have no particular advantage.

8. Conclusion

The investment banking divisions of advisor banks possess substantial private information on client firms. While barriers between the investment banking and asset management divisions of banks are intended to prevent information leakage, it is difficult to safeguard all information. To the extent that geographic proximity and/or social connections contribute to information dissemination, the asset management division of the advisor bank potentially has an advantage over other market participants. Prior literature on this subject has been inconclusive, with some papers finding evidence of such information leakage but others finding no such evidence. In contrast to these prior papers that focus on trades in the equity or debt of the target firm, we conjecture that any informed trading is more likely to occur in options markets.

Our results are consistent with the asset management divisions of advisor banks having advantages relative to other financial institutions, and with them leveraging their private information through the options markets. While many papers have utilized quarterly stock holdings as a means to estimate institutions' information sets, to the best of our knowledge, this study is the first to examine the role of the options market as a means for investment banks to leverage private information around corporate events.

Appendix

Variable Descriptions

Timing of stock and options holdings

Quarter -1: Last quarter ending prior to merger announcement, where quarters end in March, June, September, and December of each year.

Quarters -2, -3, -4, -5, -6: The quarters preceding Quarter -1.

Measures of options and stock holdings

Call Ratio: $\frac{\text{Calls}}{\text{Shrs Out}}$, number of call options held by the institution at the end of the quarter, deflated by millions of shares outstanding at the end of the quarter, winsorized at the 99.5% level. In regressions, the ratio is expressed in percentage terms.

Put Ratio: $\frac{\text{Puts}}{\text{Shrs Out}}$, defined analogously.

Net Call Ratio: $\frac{\text{Calls} - \text{Puts}}{\text{Shrs Out}}$, defined analogously.

Stock Ownership Ratio: $\frac{\text{Shares Held}}{\text{Shrs Out}}$ shares held by the institution at the end of the quarter, deflated by shares outstanding at the end of the quarter, winsorized at the 99.5% level. In regressions, the ratio is expressed in percentage terms.

Dummy variables denoting institution holding stock or options

Target Advisor: Equal to 1 if the institution served as advisor to the target firm, 0 otherwise.

Acquirer Advisor: Equal to 1 if the institution served as advisor to the acquirer firm, 0 otherwise.

Firm characteristics

Market Capitalization (abbreviated as Mkt Cap): Equal to *shares outstanding*price*, at the end of the quarter, in thousands of dollars.

Volatility: The standard deviation of daily returns over each of the three months in the quarter, averaged over the three quarters.

Option Volume: Total number of options traded in the firm over the quarter, where each unit reflects an option on 100 underlying shares.

Share Volume: Total number of shares traded in the firm over the quarter, expressed in thousands.

Stock Return: Raw stock return over the quarter, measured as the buy-and-hold return over the three months during the quarter.

Leverage: Equal to *book debt / (total assets – book equity + market value of equity)*, as of the most recent fiscal year end.

Industry Merger Activity: The percentage of firms in the same Fama-French 48-industry group that have been acquired over the past year.

EBITDA / Assets: *Operating income before depreciation / assets*, as of the most recent fiscal year end

Market to Book: Equal to *(total assets–book value of equity + market value of equity) / assets*, as of the most recent fiscal year end.

Cash / Assets: Equal to *(cash and cash equivalents) / assets*, as of the most recent fiscal year end.

CapEx / Assets: Equal to *capital expenditures / sales*, as of the most recent fiscal year end.

R&D / Sales: Equal to *R&D expenditures / sales*, as of the most recent fiscal year end.

Deal characteristics

Cash Merger: Equal to 1 if the deal was paid for entirely with cash, 0 otherwise.

Termination Fee: Equal to 1 if the target firm has to pay a termination fee if it withdraws from the merger, 0 otherwise.

Merger Rumor: Equal to 1 if there is a merger-related news story about the company during the quarter. Any news story recorded in RavenPack for which the type, category, or subtype fields is “merger,” “business-comb,” or “acquisition” is considered a merger-related news story.

Target and Acquirer in Same Industry: Equal to 1 if both the target and acquirer firms are in the same Fama-French 12-industry group.

Tender Offer: Equal to 1 if the merger is coded in SDC as being a tender offer.

Target (Acquirer) CAR_{–1,+1}: Target firm (acquirer firm) return from Day –1 through Day +1 relative to the merger announcement, minus the value-weighted market return over the same period.

Target (Acquirer) Run-up: Target firm (acquirer firm) return from the last day of Quarter –1 through Day –2 relative to the merger announcement, minus the value-weighted market return over the same period.

Premium: Market-adjusted return, measured as the firm return minus the return on the value-weighted index, over the period beginning three months prior to merger announcement and ending at the earlier of deal resolution or two months following the merger announcement.

Withdrawal: A dummy equal to 1 if the merger is withdrawn, 0 otherwise.

Institution Characteristics

High Options Institution: Equal to 1 if the institution holds an above-median number of options.

This median is defined across all holdings (not just holdings of our sample firms) and separately for hedge funds and banks.

Same Advisor Before: Equal to 1 if the institution had served as advisor in a merger or underwriter in an equity offering (IPO or SEO) for the firm within the past three years, 0 otherwise.

Poison Pill: Equal to 1 if the bank has a poison pill, 0 otherwise.

CEO-Chair Duality: Equal to 1 if the CEO is also chairman of the board, 0 otherwise.

Classified Board: Equal to 1 if the company has a classified board where only a fraction of the board is elected each year and directors serve multi-year terms, 0 otherwise.

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